

UNIVERSITY OF GREENWICH
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VerifAI: A Multimodal Deepfake Detection Framework

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Chapter 1

Abstract

Deepfakes pose a real challenge to digital trust by creating highly realistic synthetic media, which underscores the importance of developing reliable detection systems. That’s why we’re excited to introduce VerifAI—a versatile, multimodal deepfake detector that combines visual and audio cues in a comprehensive and user-friendly end-to-end system. It tackles common issues like poor cross-dataset performance, reliance on a single modality, and ease of use. By leveraging a hybrid architecture featuring MobileNetV2, ResNet18, and ViT, along with a dynamic routing mechanism, the system is designed to adapt seamlessly to various conditions. Features such as subject-level dataset splitting, clip relabeling, and multi-frame aggregation help boost its accuracy. Our testing, which included metrics like accuracy and ROC-AUC, showed strong results, though we also recognise the challenges posed by real-world distribution shifts. Additionally, explainability tools like Grad-CAM and insights from a user study confirmed that the system is both trustworthy and easy to understand. Overall, VerifAI demonstrates how combining multimodal learning with flexible design can significantly improve deepfake detection, helping to connect research advancements with real-world applications.

Declaration of AI Use

I have used AI while undertaking my assignment in the following ways:

- To develop research questions on the topic
 - To explain concepts
 - To summarise the following resources
1. Sippy, T., Enock, F.E., Bright, J. and Margetts, H.Z. (2024) survey public exposure to and concern about deepfakes in the UK, finding that while only 8% have created deepfakes, over 90% are concerned about their societal impact.
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Chapter 2

Introduction

The increasing realism of AI-generated media challenges trustworthiness, especially with deepfakes—convincing manipulated videos that are hard to detect. This project explores machine learning methods to identify such media, addressing issues like poor generalisation, limited multimodal use, and lack of interpretability. The VerifAI framework is a multimodal detection system combining visual and audio analysis using convolutional and transformer architectures to find inconsistencies in facial movements and speech. It features a dynamic inference system that switches between multimodal analysis and a visual-only fallback based on audio availability for robustness. Contributions include subject-level dataset splitting, clip-level labelling, and multi-frame inference for stability. The chapter details background, aims, methodology, and report overview to explain the system’s design and evaluation.

2.1 Background

The rapid rise of artificial intelligence has transformed digital media creation and consumption, with deepfake technology emerging as a disruptive innovation. Deepfakes, synthetic media created through deep learning, can convincingly imitate real people. While they have legitimate uses in entertainment, they pose risks to trust, accuracy, and security. Easier access to deepfake tools enables malicious activities like misinformation and identity theft. These issues impact areas like politics, journalism, and law, where authenticity matters. Therefore, automatic detection of manipulated media has become urgent. Early detection methods used convolutional neural networks to identify visual artefacts with high accuracy, but faced limitations such as poor generalisation, reliance on visual data, and lack of interpretability. These issues reduce trust and practicality, especially outside controlled environments. Recent research emphasises the need for robust, explainable detection systems adaptable to real-world conditions Lyu (2020) Heidari et al. (2023). Industry solutions like content moderation and metadata verification exist, but are limited by reliance on controlled data. This project aims to overcome these limitations by improving generalisation with multi-dataset training, integrating multimodal analysis, and creating resilient systems, combining machine learning and software engineering to deploy effective real-world deepfake detection.

2.2 Aims and Objectives

This project aims to develop and evaluate a multimodal deepfake detection framework to classify real or fake videos in real-world conditions. It focuses on objectives aligned with the project proposal (see Appendix A), providing measurable criteria for success in later chapters.

1. This study explores the field of deepfake detection, emphasising key challenges such as generalisation, multimodal modelling, and interpretability.
2. Design and implement a comprehensive data preprocessing pipeline that can handle heterogeneous datasets and extract visual and audio features in a reproducible manner.
3. To create a multimodal fusion approach that integrates visual and audio features at the feature level, with the goal of improving detection accuracy.
4. Implement a dynamic inference mechanism that automatically chooses between multimodal and visual-only detection depending on input characteristics.
5. To conduct a thorough system evaluation, utilise quantitative performance metrics, perform cross-dataset testing, validate in real-world scenarios, and deploy as a functional API.

2.3 Methodology

This project employs a design science and experimental research methodology, combining system development with performance evaluation. Design science research, common in software engineering, supports iterative refinement and empirical validation. The process is iterative and modular, allowing components to be designed, tested, and improved independently. This approach was chosen over a linear process to enable experimentation with models and incremental improvements. A development timeline was established to manage the project and ensure progress, as shown in Figure 2.1.

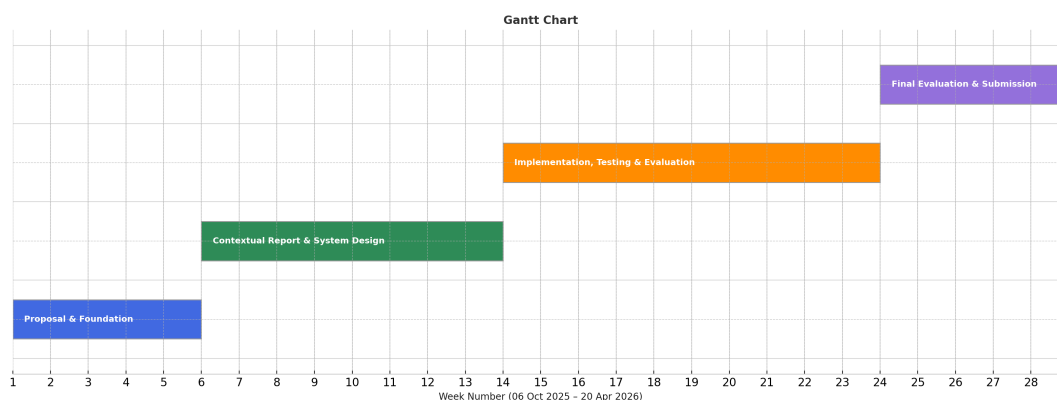


Figure 2.1: A project timeline showing the phased development of the VerifAI system, from initial proposal through to final evaluation.

The methodology employed in this project comprised several interconnected stages, each designed to support both experimental investigation and practical system development. Initially, a comprehensive literature review and problem analysis were conducted to identify key limitations in existing deepfake detection approaches and to inform future design decisions. This was followed by data collection and preprocessing, during which multiple datasets were combined using a structured pipeline that included frame extraction, audio extraction, MFCC feature generation, subject-level dataset splitting, and clip-level authenticity relabelling. Model development and experimentation then proceeded to establish baseline performance and explore alternative architectures, such as convolutional and transformer-based models. Building on these results, a multimodal fusion strategy was formulated using feature-level integration of visual and audio embeddings via a neural network-based fusion mechanism, enabling the system to learn cross-modal relationships. The final system was subsequently integrated into a FastAPI-based backend to facilitate real-time inference through a REST interface. Finally, the system underwent assessment through a combination of quantitative performance metrics, cross-dataset validation, explainability analysis, and a user study focusing on usability and trust. This methodology was chosen to strike a balance between technical experimentation and practical deployment considerations. The iterative approach allowed early identification of model limitations and guided the selection of more robust and adaptable architectures in subsequent stages.

2.4 Report Roadmap

This report guides the development and evaluation of the VerifAI system. It reviews existing deepfake detection techniques, their limitations, and establishes a theoretical basis. It covers requirements analysis, system design, and implementation, including preprocessing, model architectures, and deployment. The system’s performance is evaluated with metrics and assessments like explainability and user trust. The report discusses legal, social, ethical, and professional implications, reflecting on broader impact and responsibilities. It concludes with a summary, success, and future research directions.

Chapter 3

Literature Review

This chapter provides a comprehensive review of deepfake detection research by synthesising key themes such as detection techniques, dataset challenges, multimodal methodologies, and explainability. Its objective is to establish the current state of knowledge, identify existing gaps, and justify the rationale behind design choices. The review emphasises ongoing challenges related to generalisation, multimodal integration, and real-world application, which influence the VerifAI framework.

3.1 Introduction

Deepfake detection has become a crucial research area due to the rapid advancement of generative artificial intelligence. While early research primarily aimed to improve detection accuracy, recent efforts increasingly emphasise robustness, interpretability, and real-world applicability. This shift reflects a growing understanding that detection systems must operate reliably outside controlled experimental environments. A significant body of literature has explored the technical challenges associated with deepfake detection. Comprehensive surveys, such as Heidari et al. (2023), highlight that although deep learning methods attain high performance on benchmark datasets, they frequently encounter difficulties in generalising to unseen data and adapting to evolving manipulation techniques. Similarly, Lyu (2020) identifies key limitations, including dataset bias, susceptibility to adversarial attacks, and the lack of standardised evaluation protocols. Concurrently, research has commenced examining the broader societal impacts of deepfakes. Empirical studies reveal that public concern regarding deepfake technologies is significant, yet users' detection capabilities remain limited, thereby increasing reliance on automated detection systems Sippy et al. (2024). This emphasises the need to develop solutions that are not only accurate but also interpretable and trustworthy. Based on these insights, this chapter reviews the literature organised around four principal themes: the development of detection techniques, limitations related to datasets and generalisation, multimodal detection methods, and explainability and trust. These themes underpin the design choices discussed in this project.

3.2 Evolution of Deepfake Detection Techniques

Early deepfake detection methods primarily relied on convolutional neural networks (CNNs) designed to identify spatial artefacts in manipulated images and video frames.

These artefacts include inconsistencies in lighting, texture, and facial blending that manifest during synthesis. Empirical evaluations of architectures such as VGG, ResNet, and DenseNet demonstrate high accuracy—often exceeding 90%—on controlled benchmark datasets like FaceForensics++ Rössler et al. (2019) and other curated datasets Ahmed et al. (2022), Heidari et al. (2023). However, subsequent research has indicated that such performance is frequently overstated due to overfitting to dataset-specific artefacts. Models trained on limited datasets typically encounter difficulties in generalising to more complex, real-world scenarios where variations in compression, resolution, and manipulation techniques are considerably greater Lyu (2020), Zi et al. (2020). This discrepancy underscores a fundamental challenge in deepfake detection: the disparity between benchmark success and real-world robustness. To address these issues, research has increasingly shifted towards architectures capable of modelling temporal and contextual dependencies. Transformer-based approaches incorporate attention mechanisms that model long-range spatial and temporal relationships across video frames, thereby enhancing the detection of subtle motion and facial dynamics inconsistencies Gong & Li (2024). Hybrid architectures that combine CNN-based feature extraction with transformer-based sequence modelling further improve performance by integrating local spatial sensitivity with a comprehensive understanding of global context Heidari et al. (2023). While CNN-based methods remain computationally efficient, transformer models offer superior contextual modelling at the expense of increased computational demands, illustrating a trade-off between performance and scalability. Despite these advances, the literature consistently emphasises that no single architecture provides a comprehensive solution. CNN models are advantageous for their computational efficiency and suitability for spatial analysis, whereas transformer models possess enhanced representational capacity but require greater resources and are more challenging to train Gong & Li (2024). This trade-off suggests that practical detection systems should prioritise robustness, efficiency, and generalisation over architectural complexity. Recently, research has focused on ensemble learning and adversarially robust training methods designed to mitigate vulnerabilities in individual models. These strategies aim to enhance resilience against evolving manipulation techniques and adversarial attacks, reflecting a shift from static classifiers to adaptive and more generalisable detection systems Gong & Li (2024). Furthermore, while early multimodal systems predominantly employed late fusion strategies to combine modality-specific predictions, recent investigations have explored feature-level (deep) fusion approaches. These enable models to learn cross-modal interactions directly from integrated representations, thereby improving the detection of subtle inconsistencies between modalities Gong & Li (2024).

3.3 Dataset Challenges and Generalisation

The effectiveness of deepfake detection systems fundamentally depends on the quality and diversity of datasets used for training and evaluation. Early benchmark datasets, such as FaceForensics++ Rössler et al. (2019), played a crucial role in establishing baseline performance; however, they were created under controlled conditions with limited manipulation techniques. Consequently, models trained on these datasets often achieved high accuracy but generalised poorly to real-world data Lyu (2020). This gap highlights a key limitation in early research: reliance on curated datasets that do not adequately reflect real-world variability. In response, later datasets have aimed to introduce greater realism and diversity. For example, WildDeepfake Zi et al. (2020) focuses on in-the-wild forgeries

collected from online platforms, while KoDF Kwon et al. (2021) includes demographic diversity to reduce bias in detection systems. Empirical studies using these datasets reveal that cross-dataset performance can decline significantly when models encounter unfamiliar data distributions, emphasising the limitations of dataset-specific learning Gong & Li (2024). The addition of multimodal datasets like FakeAVCeleb Khalid et al. (2021) marks a further step forward by incorporating both visual and audio manipulations. Research leveraging these datasets indicates that multimodal approaches can substantially improve detection performance by leveraging complementary information across modalities Heidari et al. (2023). This evolution reflects a wider move towards more comprehensive detection frameworks that extend beyond visual analysis alone. Apart from dataset diversity, recent research underscores the importance of data splitting strategies and task formulation. Incorrect splitting at the frame or clip level can cause identity leakage, leading to the same subjects appearing in both the training and test sets, artificially boosting performance metrics. Subject-level splitting has therefore been suggested as a more realistic evaluation approach. Similarly, redefining the classification task at the clip level-rather than relying on modality-specific labels-better aligns with real-world detection scenarios. Despite these advances, the literature consistently stresses that dataset diversity alone does not fully solve the generalisation challenge. Many datasets still fail to capture the full complexity of real-world conditions, including variations in compression artefacts, lighting, recording devices, and post-processing pipelines. As a result, a substantial gap remains between experimental results and practical deployment, highlighting the need for more robust evaluation protocols and models designed to improve generalisation Lyu (2020).

3.4 Multimodal Detection and Explainability

A significant trend in recent deepfake detection research is the shift from unimodal to multimodal approaches. Traditional systems that rely solely on visual information are inherently limited because they cannot detect inconsistencies across different data modalities. In contrast, multimodal frameworks combine complementary sources of evidence, such as audio and video signals, enabling the detection of cross-modal discrepancies that are considerably more difficult to forge convincingly Heidari et al. (2023). Empirical studies using multimodal datasets demonstrate that fusing audio and visual features can notably enhance both detection accuracy and robustness. For example, experiments on the FakeAVCeleb dataset reveal that multimodal fusion strategies consistently outperform unimodal baselines by leveraging complementary information across modalities Khalid et al. (2021). However, integrating heterogeneous data sources poses additional challenges, especially regarding temporal alignment, feature representation, and modality weighting, which remain open research issues Gong & Li (2024). Recent work has also investigated adaptive multimodal systems capable of functioning under varying input conditions, including scenarios where one modality may be absent or unreliable. Such approaches are particularly relevant for real-world deployment, where data quality cannot be assured, and rigid multimodal assumptions may detract from system usability and resilience. Alongside improvements in detection performance, explainability has become a vital requirement for practical implementation. Many state-of-the-art deep learning models operate as black boxes, producing predictions without revealing insight into the decision-making process. This lack of interpretability not only limits user trust but also

raises significant concerns regarding accountability and reliability in high-stakes applications Lyu (2020). To tackle these challenges, a variety of explainability techniques have been proposed. Methods such as Grad-CAM visualisations and attention mapping aim to enhance interpretability by highlighting the spatial and temporal regions that most strongly influence model predictions Heidari et al. (2023). More advanced methods extend explainability through forensic analysis and provenance tracking, allowing users to validate authenticity based on multiple complementary signals Gong & Li (2024). Despite these advances, explainability remains insufficiently incorporated into real-world systems. Existing approaches tend to prioritise performance optimisation over user-centred design, resulting in limited transparency during deployment. This gap underscores the need for detection frameworks that effectively balance accuracy, robustness, and interpretability.

3.5 Conclusion

This chapter reviews deepfake detection research, noting progress, challenges, and gaps. Despite better accuracy, issues like poor dataset generalisation, limited multimodal use due to complexity, and lack of explainability remain. These gaps highlight the divide between academic success and real-world application, underscoring the need for adaptive, multimodal solutions like VerifAI. This system employs multi-dataset training, feature-level multimodal fusion, and adaptive inference for robustness, focusing on subject-level evaluation, realistic tasks, and deployment to connect research and practice. The next chapter will examine how these insights offer a comparative review of existing real-world products.

Chapter 4

Product Review

4.1 Introduction

Deepfake detection has been extensively examined, resulting in numerous systems with varying complexity, performance, and usability. Many remain experimental or proprietary, often lacking transparency and adaptability. This chapter reviews key detection systems to highlight their strengths and weaknesses relative to the project. The evaluation covers generalisation, multimodal integration, explainability, and real-world use, focusing on gaps like cross-dataset performance, multimodal reasoning, and user trust. This analysis helps refine the VerifAI framework, especially regarding multimodal fusion and adaptive inference.

4.2 Product 1: CNN-Based Deepfake Detection Systems

4.2.1 Overview

CNN-based deepfake detection systems, some of the earliest and most used methods, leverage convolutional neural networks to identify visual artefacts in manipulated media, such as texture, lighting, and facial discrepancies. Architectures like Xception, ResNet, and VGG perform well on datasets like FaceForensics++, but their effectiveness is limited outside controlled environments.

4.2.2 Generalisation Capability

CNN-based detectors often struggle to generalise, performing well on training data but faltering on new, unseen data due to overfitting to dataset-specific artefacts like compression or generation methods. This limits their ability to detect new deepfakes or videos in different conditions.

4.2.3 Multimodal Integration

CNN-based systems are limited by their unimodal design, restricting their ability to detect cross-modal inconsistencies common in deepfake attacks. This lack of multimodal

reasoning is especially problematic when manipulations occur in audio or when cross-modal clues are key indicators of forgery.

4.2.4 Explainability and Usability

Most CNN-based detection systems are opaque, providing binary predictions without explanations. Although some use visualisation like heatmaps, these are often unsuitable for non-experts. This lack of interpretability diminishes user trust and limits deployment, especially in high-stakes settings.

4.3 Product 2: Multimodal Deepfake Detection Systems (FakeAVCeleb-based)

4.3.1 Overview

Multimodal deepfake detection systems improve accuracy by combining visual and auditory data. The FakeAVCeleb dataset advances this by providing paired audio-visual manipulations. These systems usually use CNN visual encoders and audio classifiers with fusion mechanisms for prediction.

4.3.2 Generalisation Capability

Multimodal systems demonstrate greater robustness than unimodal methods, especially in detecting discrepancies between modalities. However, they still face challenges in generalising across different datasets, particularly when trained on limited multimodal data resources. In practice, their effectiveness often relies on dataset-specific distributions and may decline in real-world scenarios.

4.3.3 Multimodal Integration

These systems' main advantage is integrating multiple information sources. Multimodal detectors analyse auditory and visual signals, spotting inconsistencies missed by single modalities. Many use late fusion, combining predictions at the decision level, which limits learning complex interactions. Recent advances favour feature-level (deep) fusion, integrating embeddings before classification, improving cross-modal understanding, but its complexity restricts use. Most systems assume both modalities are present, reducing effectiveness when audio is missing or poor.

4.3.4 Explainability and Usability

While multimodal systems boost detection, they often add complexity and reduce interpretability. Fusion mechanisms can hide the role of individual modalities, making decisions harder to understand. Most remain research prototypes and aren't user-facing, limiting practical use. Compared to CNN systems, multimodal approaches offer more robustness through cross-modal analysis but add complexity and rely on data.

4.4 Product 3: Microsoft Video Authenticator

4.4.1 Overview

Microsoft Video Authenticator, developed by Microsoft, detects deepfakes by analysing images and videos for artificial manipulation. It uses AI to spot subtle artefacts like blending issues and unnatural transitions. Unlike research models, it is designed for deployment in media verification and misinformation mitigation.

4.4.2 Generalisation Capability

Though trained on diverse data, its performance and generalisation remain undisclosed, complicating effectiveness assessment on unseen data. This underscores a common industry limitation: strong deployment with limited transparency and reproducibility.

4.4.3 Multimodal Integration

The system primarily emphasises visual analysis and lacks audio detection, risking missed mismatches between audio and video. This reveals a literature gap: many tools stay unimodal despite the growth of multimodal deepfakes.

4.4.4 Explainability and Usability

Microsoft Video Authenticator has an easy interface and provides a confidence score for manipulation, making it accessible to non-experts. However, it lacks detailed explanations like feature attribution or heatmaps, which could reduce trust in high-stakes contexts.

4.4.5 Real-World Applicability

This tool’s main advantage is its quick deployment, designed for practical use like journalism and content verification. However, its limitations in multimodality, transparency, and interpretability show that improvements are needed.

4.5 Key findings

The comparative analysis provides essential insights into the strengths and weaknesses of current deepfake detection techniques.

Criterion	CNN-Based	Multimodal	Microsoft	VerifAI
Generalisation	Low	Moderate	Unknown	High
Multimodal Capability	No	Yes (limited)	No	Adaptive
Fusion Strategy	N/A	Late fusion	N/A	Deep fusion
Handling Missing Modalities	No	No	No	Yes
Explainability	Low	Low	Moderate	High
Real-World Usability	Low	Moderate	High	High

Table 4.1: Comparison of deepfake detection approaches

This comparison highlights key limitations: 1. CNN methods lack robustness and struggle to generalise to new data. 2. Multimodal frameworks improve detection but are often rigid and computationally intensive. 3. Industry tools are user-friendly but often lack transparency and multimodal reasoning. These findings show a trade-off: high-performance approaches sacrifice flexibility and usability, while deployment tools lack transparency and depth. This motivates VerifAI’s hybrid, adaptable design.

In contrast, the proposed VerifAI system mitigates these limitations through:

- Feature-level multimodal fusion, enabling more effective cross-modal representation learning.
- Dynamic inference routing, ensuring dependable operation across diverse input scenarios.
- A balanced approach that amalgamates performance, interpretability, and usability for pragmatic deployment.

4.6 Conclusion

This chapter examines current deepfake detection systems, highlighting their strengths and limitations across key dimensions. Despite advances, issues like poor generalisation, inflexible multimodal assumptions, and a lack of focus on user trust hinder progress. These findings support developing VerifAI, which uses deep multimodal fusion, adaptive inference, and a deployment-focused design to address these gaps. The next chapter discusses requirements and system design, turning limitations into a coherent, adaptive framework bridging research and application.

Chapter 5

Requirements Analysis and Design

This chapter’s requirements analysis delineates the functional and non-functional expectations for the VerifAI system, elucidating its fundamental capabilities and limitations. These requirements establish a structured foundation for the design process, ensuring that decisions are aligned with the overarching objectives of the project. Subsequently, the chapter converts these requirements into a concrete, modular, and adaptable system design that addresses aspects such as generalisation, multimodal integration, and real-world applicability. It provides justifications for key design decisions, evaluates alternative options, and demonstrates how the architecture incorporates both insights and empirical findings.

5.1 Introduction

This chapter encompasses the requirements analysis and design of the VerifAI system, transforming the problem outlined in preceding chapters into a clear and implementable solution. Its objective is to delineate the system’s functional and non-functional requirements, substantiate the design decisions, and demonstrate the development of the final architecture through systematic analysis, testing, and refinement. The chapter commences with an explanation of how the system requirements were established, drawing on the literature review, product analysis, and experimental results. Subsequently, it details the primary functional and non-functional requirements, including their respective priorities. The chapter then describes the overall system design, encompassing architecture, components, and user interface considerations. Finally, it discusses the rationale behind the design choices and their alignment with the overarching project objectives.

5.2 Requirements Analysis

5.2.1 Requirements Elicitation and Analysis

The VerifAI system requirements stemmed from multiple sources to ensure it addresses technical challenges and usability. The literature review revealed limitations in existing deepfake detectors, like poor generalisation, multimodal assumptions, and limited explainability, highlighting the need for a system resilient across datasets, integrating multiple modalities, and adaptable in real-world conditions. The product review noted current systems often prioritise either academic performance or practicality, rarely both;

most multimodal systems assume all modalities are available, while industry tools lack multimodal reasoning and transparency. This underscored the need for high performance, adaptability, and user-friendliness. Early experiments showed that single-model architectures are insufficient, unimodal approaches can't detect cross-modal inconsistencies, and static pipelines fail under varied inputs. These insights led to a hybrid, adaptive inference multimodal system. User evaluations emphasised clear, interpretable outputs, transparency, and minimal interactions. These findings demonstrate deepfake detection requires a system combining multimodal reasoning, adaptive inference, and deployment constraints, shaping VerifAI as a modular, hybrid framework rather than a single model.

5.2.2 Assumptions and Constraints

Several assumptions and constraints define the project's scope and feasibility. Assumptions include: (a) Input videos are of sufficient quality for frame extraction. (b) Audio can be reliably extracted and processed. (c) Users interact via a web interface. (d) Deepfake manipulations show detectable inconsistencies. Constraints include: (a) Limited computational resources affecting model choice and training. (b) Dataset limitations don't reflect real-world variability. (c) Focus on core features due to time constraints. (d) Designed for API deployment rather than large-scale infrastructure.

5.2.3 Functional Requirements

The functional requirements delineate the fundamental capabilities of the system and are prioritised using the MoSCoW framework.

Must-Have

- Accept video input and validate format, size, and integrity.
- Extract and synchronise visual frames with audio signals.
- Detect the presence and quality of audio for inference routing.
- Use multimodal deepfake detection through feature-level fusion.
- Dynamically route inference between hybrid and fallback models.
- Aggregate predictions from multiple frames to generate stable clip-level results.
- Return structured predictions with label, probability, and inference mode.

Should-Have

- Provide clear and well-structured output.
- Support multiple video formats (.mp4, .mov, .avi).
- Enable consistent and stable predictions.

Could-Have

- Provide visual explainability mechanisms.
- Store and retrieve previous analysis results.
- Include extended metadata analysis.

Won't-Have (for now)

- Real-time streaming detection.
- Full forensic localisation or manipulation tracing.

Justification This prioritisation ensures essential detection within constraints, allowing future improvements. Core capabilities maintain system viability, while features like explainability and analytics are secondary. Figure 5.1 shows VerifAI’s use case, providing a high-level user interaction overview.

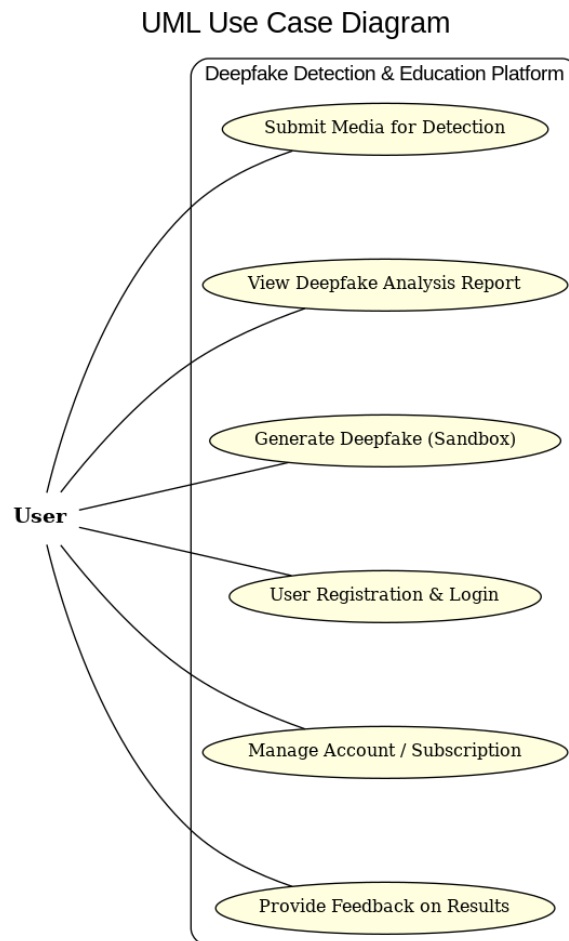


Figure 5.1: Use case diagram illustrating user interactions with the VerifAI system.

5.2.4 Non-functional Requirements

Non-functional requirements delineate the quality attributes and operational characteristics of the system.

Must-Have

- High detection accuracy with consistent performance across classes.
- Resilience to variations across datasets and input conditions.
- Reliable and consistent management of API responses.

Should-Have

- Low inference latency to enable efficient processing.
- Scalability of the API to accommodate increased usage.
- An intuitive interface designed for users without technical expertise.

Could-Have

- Advanced visualisation capabilities to improve result interpretation.
- Performance optimisation for large-scale or production deployment.

Justification The prioritisation emphasises robust performance and system reliability, crucial for deepfake detection. While additional features like better visualisation and scalability are helpful, they are secondary to accurate detection. Usability is also vital for real-world deployment.

5.3 Design

5.3.1 Design Overview

The VerifAI system employs a modular, adaptable architecture to surpass current deepfake detection limits. Instead of static models, it combines multiple components in a flexible pipeline. Developed through iterative testing, initial unimodal or static versions were replaced with a hybrid approach due to challenges with generalisation, modality management, and robustness.

5.3.2 System Architecture

The final system architecture consists of the following layers:

Input Processing Layer Handles user video uploads, extracts frames for analysis, and processes audio for examination.

Modality Detection Layer Verifies a valid audio signal and manages inference routing.

Model Inference Layer Visual encoder: MobileNetV2 (multimodal) / ViT (fallback); audio encoder: ResNet18 with MFCC; fusion model: combines visual and audio embeddings using an MLP.

Decision Layer Utilises multimodal embeddings with a learned fusion mechanism and produces classification through an MLP decision head.

Output Layer Returns the final label, shows prediction probability, and indicates inference mode.

Design Rationale Feature-level fusion enhances cross-modal relationships, while dynamic inference routing adapts to input variability, boosting robustness. Independent encoders enable separate optimisation, multi-frame inference stabilises predictions, and a modular design supports scalability and updates. The VerifAI detection pipeline overview in Figure 5.2 shows input flow through preprocessing, inference, and decision components, with internal structure in Figure 5.3.

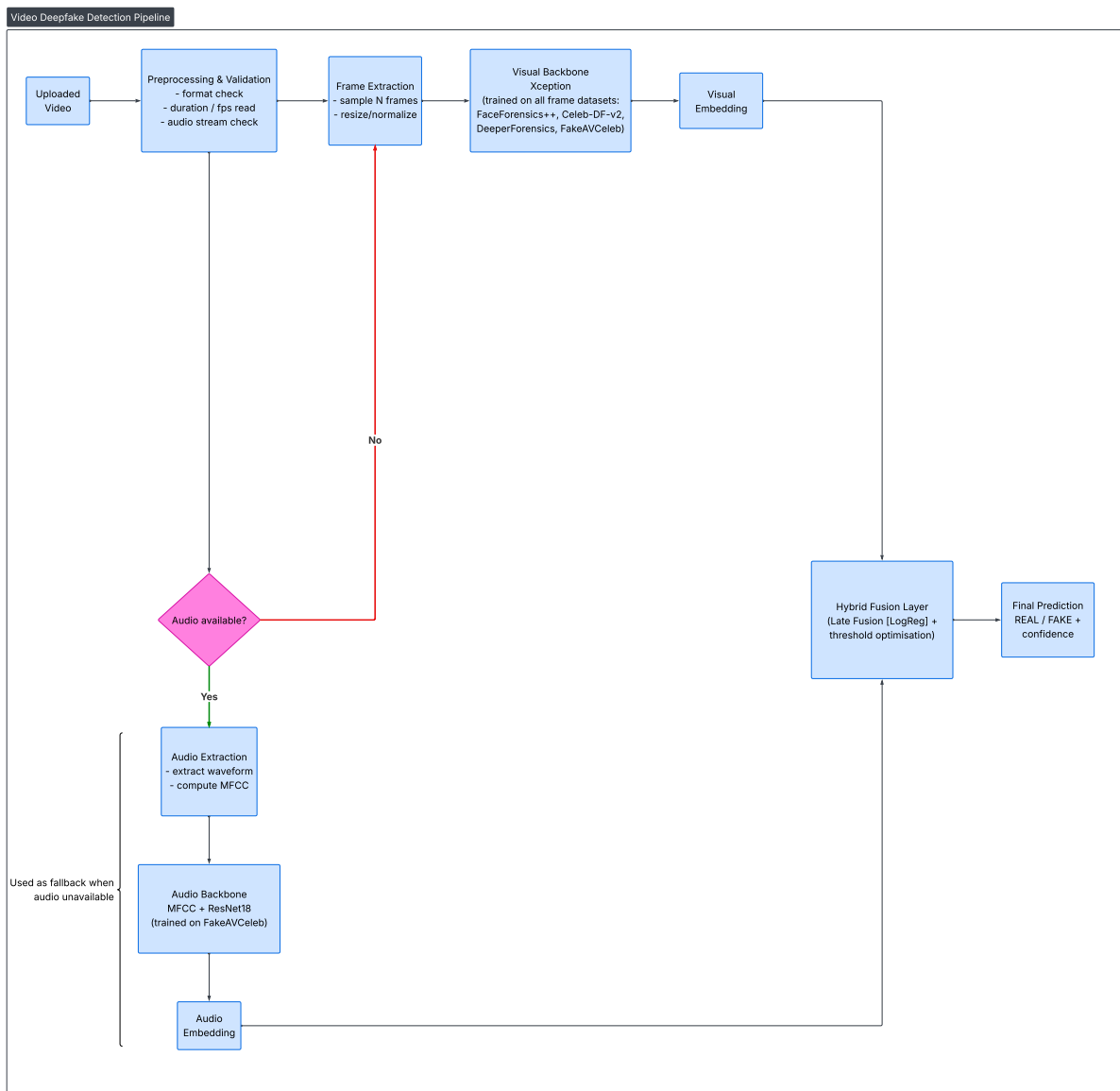


Figure 5.2: End-to-end deepfake detection pipeline of VerifAI, including preprocessing, multimodal inference, and dynamic routing.

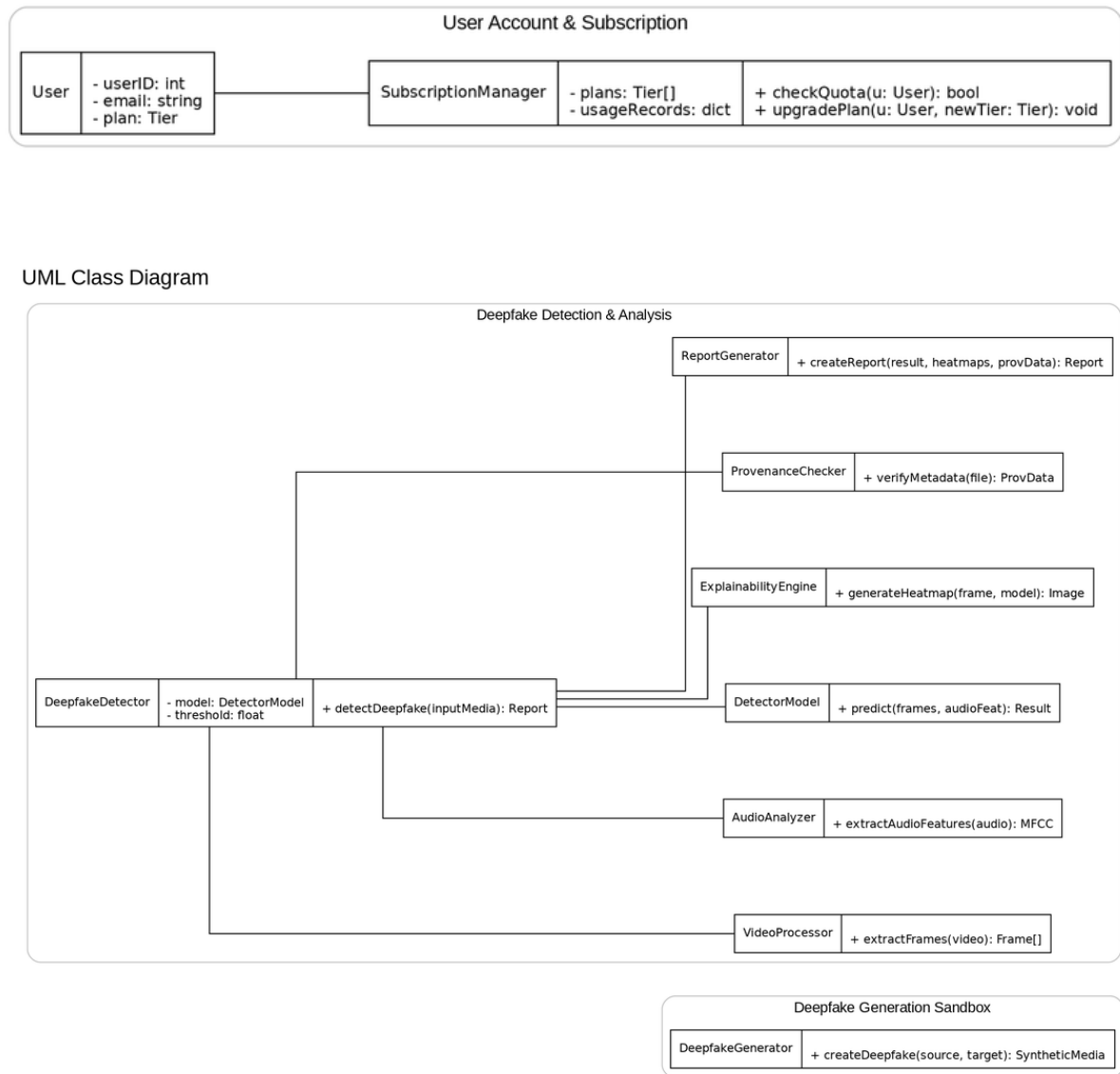


Figure 5.3: UML class diagram representing the internal structure and component relationships of the VerifAI system.

5.3.3 UI/UX Design

The system incorporates a web-based user interface engineered to emphasise usability, simplicity, and accessibility for a diverse user demographic.

Key Features A single-step workflow facilitates seamless upload and detection, provides real-time progress updates via a visual progress indicator, delivers structured JSON output for detailed and transparent result presentation, and features a clean, minimal interface designed to minimise cognitive load.

Design Evolution The initial implementation employed a multi-step workflow, resulting in inefficiencies and increased user effort. The final design adopts a streamlined interaction model, significantly improving usability and responsiveness.

Rationale Minimise user friction and interaction complexity, enhance trust through transparent and interpretable outputs, and ensure accessibility for non-technical users.

5.3.4 Design Decisions and Trade-offs

Several key trade-offs were assessed during the system design process, as summarised in Table 5.1.

Decision	Trade-off
ViT fallback vs CNN-only	Robustness vs simplicity
Deep fusion vs late fusion	Performance vs interpretability
Dynamic routing vs static pipeline	Flexibility vs complexity
Multimodal vs unimodal	Accuracy vs dependency on data

Table 5.1: Key design trade-offs

These trade-offs exemplify a conscious prioritisation of robustness and deployability over purely theoretical performance metrics. Notably, the decision to implement dynamic routing and deep fusion represents a transition from a model-centric to a system-centric engineering approach.

5.4 Conclusions

This chapter outlines the VerifAI system’s requirements and design, showing how the problem domain and literature findings were turned into a practical solution. It highlights the need for a system that is robust, adaptable, and capable of multimodal reasoning. The design employs a modular architecture with deep multimodal fusion and dynamic inference routing. These choices shape the implementation discussed in the next chapter, where the architecture, fusion process, and routing mechanism are realised. The next chapter details the system’s implementation.

Chapter 6

Implementation

This chapter expounds on the implementation of the VerifAI system, emphasising key technical facets such as data preprocessing, model development, multimodal fusion, inference management, deployment, and evaluation. It underscores the design decisions that distinguish it from more basic deepfake detection tools and addresses development challenges encountered. These sections illustrate the progression of the project from conceptualisation to a deployable, empirically validated prototype.

6.1 Introduction

This chapter describes the development of the final VerifAI prototype, showing how requirements and design decisions from the previous chapter were turned into a deepfake detection system. It highlights the system’s technical architecture, key engineering choices, and deployment methods. Rather than full code details, it focuses on critical parts: preprocessing, visual and audio processing, multimodal fusion, inference, API deployment, and testing. Special features that set the system apart include dynamic modality-aware inference, subject-level data handling, clip-level relabeling, and multi-frame aggregation for stability. This chapter is vital because the project’s value relies on both model accuracy and a clear, reproducible architecture suited for deployment. It covers both machine learning and software engineering aspects of integrating the prototype into a backend service.

6.2 Code Repository Outline

The VerifAI system was constructed with a modular architecture, ensuring a clear delineation among data processing, model development, and deployment components. The codebase was primarily developed from scratch, aside from standard deep learning libraries and pre-trained model backbones.

High-Level Structure

Data Module (data/, ml/ & pipelines/) Data loading and management, extracting frames from videos, extracting and preprocessing audio, and generating MFCC features.

Model Module (ml/ & pipelines/) The visual models include MobileNetV2, ViT, and Xception; the audio model uses ResNet18; and the fusion model employs an advanced deep multimodal architecture at the feature level, not probability-level. Visual frames are encoded with MobileNetV2 for embeddings, and audio is converted to MFCC features processed via ResNet18. The modality-specific embeddings are integrated into a comprehensive representation, then fed to an MLP for cross-modal learning and classification. This approach models relationships directly between audio and visual features, improving detection of subtle cross-modal inconsistencies.

Training Module (ml/ & pipelines/) Training and optimisation scripts, evaluation pipelines, and performance metric calculations.

API Module (backend/) Implementation of inference pipeline, request, and response management in a FastAPI backend.

Utilities Module (venv/) Preprocessing and auxiliary functions, data augmentation methodologies, and utilities for logging and monitoring.

Libraries Used PyTorch is utilised for model development and training, OpenCV for the extraction of video frames, Librosa for audio processing, FastAPI for backend deployment, and NumPy, along with Pandas for data manipulation and analysis.

Justification The modular design provides several significant benefits: the reusability of individual components across various stages of the pipeline, enhanced maintainability and simplified debugging processes, and scalability that facilitates future extensions and the incorporation of additional models or features.

6.3 Data Preprocessing Pipeline

A crucial component of the system implementation is the preprocessing pipeline, which is engineered to handle heterogeneous datasets and to prepare inputs for both visual and audio models consistently and reliably.

Pipeline Stages

Dataset Integration Integration of multiple datasets, including FaceForensics++, Celeb-DF, DeeperForensics, and FakeAVCeleb, along with the standardisation of data formats to ensure consistency across training inputs.

Frame Extraction Extraction of representative frames from video sequences; standardisation of image resolution and format for model compatibility.

Audio Extraction Extraction of audio streams from video inputs, where available, and the conversion of audio data into a standardised format for subsequent processing.

Feature Extraction Generation of Mel-Frequency Cepstral Coefficients (MFCC) features for audio-based modelling, and normalisation of visual inputs to enhance training stability.

Data Augmentation Using random cropping, horizontal flipping, colour jitter, and random erasing to augment data and improve robustness.

Design Rationale

The preprocessing pipeline aimed to enhance generalisation across datasets, minimise overfitting to dataset-specific artefacts, and enable robust multimodal learning. It used subject-level dataset splitting to prevent identity leakage between training and evaluation, ensuring performance reflects generalisation. Additionally, clip-level authenticity relabelling aligned classification with real-world deployment, focusing on whether a video is authentic regardless of manipulation.

6.4 Model Development

Visual Models

The development process involved evaluating visual architectures such as MobileNetV2, ViT, and Xception, revealing issues with stability and generalisation, especially in early transformer models (Figure 6.1). Instead of relying on a single architecture, the final system uses a dual-role design: a MobileNetV2 encoder for efficient visual embeddings integrated into a multimodal pipeline, and a ViT as a visual fallback when audio data is unavailable or unsuitable. This ensures robust visual capabilities and resilience to input variations, crucial for real-world applications where audio may be missing or corrupted. The approach shifts from single-model optimisation to system robustness and adaptability. As shown in Figure 6.2, the visual model achieves stable convergence and strong discriminative performance across datasets.

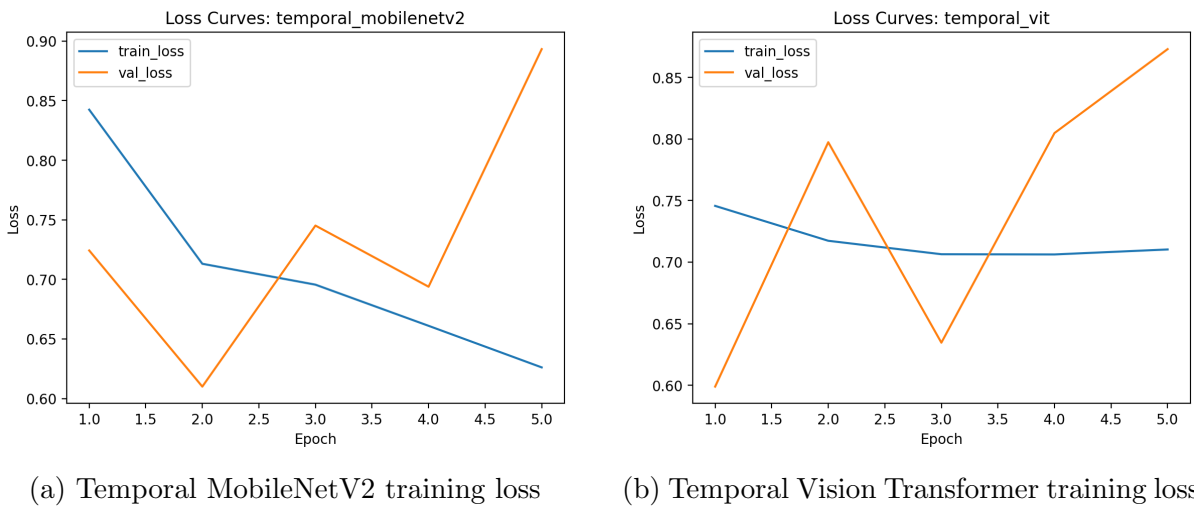
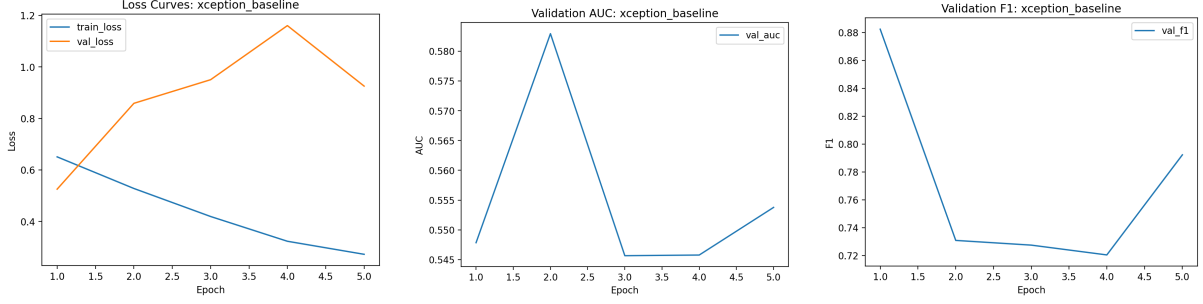


Figure 6.1: Training loss curves for early temporal and transformer-based visual models, illustrating instability and convergence challenges.



(a) Training & validation loss

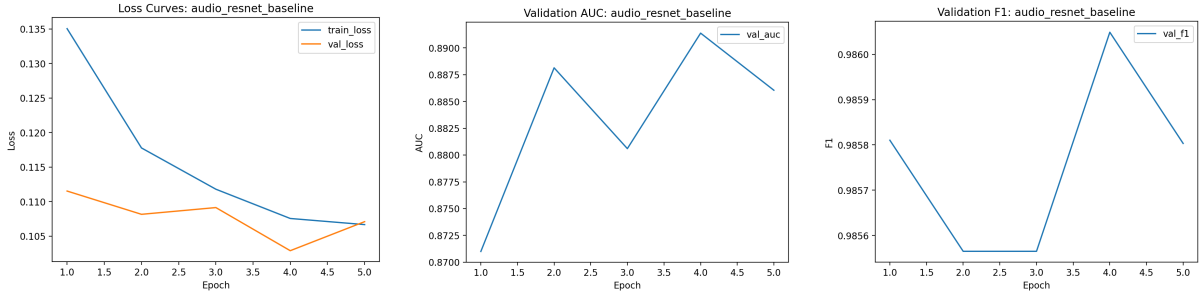
(b) Validation ROC-AUC

(c) Validation F1-score

Figure 6.2: Training and validation performance of the visual model across epochs.

Audio Model

The audio component used MFCC feature extraction with a ResNet18 classifier. Alternative methods, like raw waveform models, had limitations such as sensitivity to class imbalance and poor generalisation. The MFCC approach was chosen for its stability, consistency, and interpretability. Figure 6.3 shows the training and validation performance, demonstrating stable convergence and effective classification.



(a) Training & validation loss

(b) ROC-AUC over epochs

(c) F1-score over epochs

Figure 6.3: Training and validation metrics of the ResNet18 audio model, encompassing loss, ROC-AUC, and F1-score across epochs.

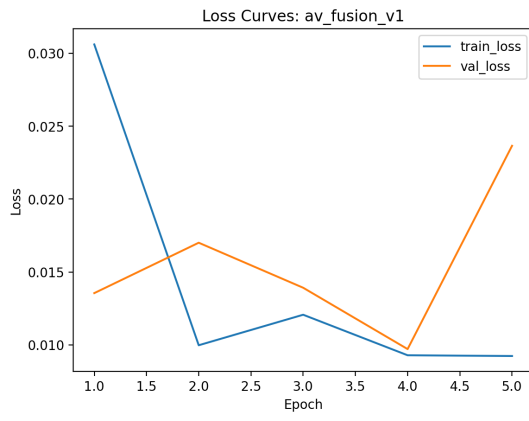
Multimodal Fusion

The ultimate system employs a late fusion methodology to integrate predictions from both the visual and audio models.

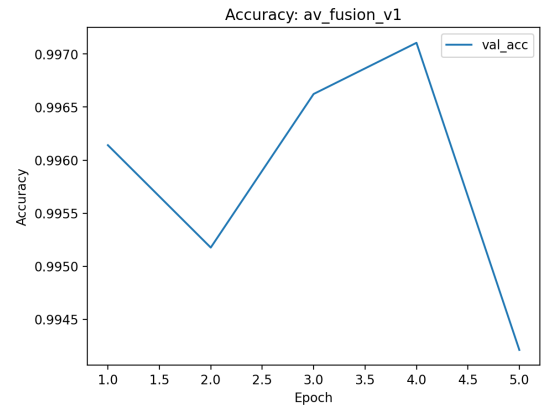
Inputs Visual prediction probability and audio prediction probability.

Fusion Model Logistic regression classifier.

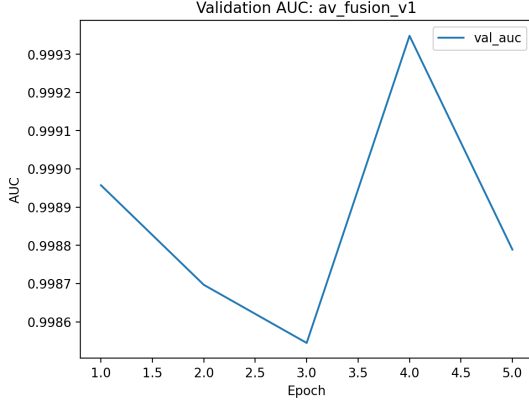
Rationale The late fusion approach was chosen for its lower overfitting risk, better interpretability, and flexible tuning of visual and audio models. Figure 6.4 shows the training dynamics, with superior performance over unimodal methods and stable convergence.



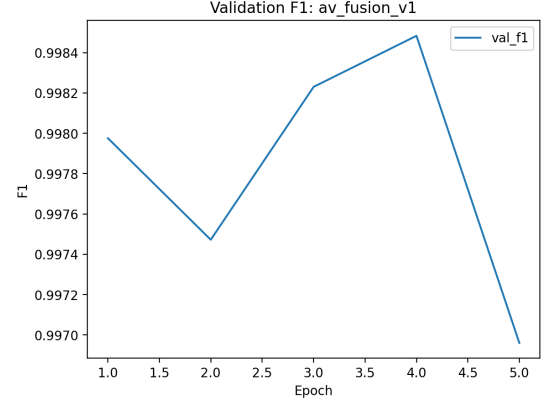
(a) Training and validation loss



(b) Accuracy over epochs



(c) Validation ROC-AUC



(d) Validation F1-score

Figure 6.4: Training and validation performance metrics of the multimodal fusion model, encompassing loss, accuracy, ROC-AUC, and F1-score across epochs.

6.5 System Integration and Deployment

The system, built with FastAPI, enables real-time inference via a scalable API. Users upload videos, which are preprocessed to extract frames and audio. It checks for a valid audio stream and selects the inference mode: hybrid with both audio and visual models, or visual-only if audio is missing. The system displays classification, prediction probability, and inference mode. Its key innovation is dynamic inference, allowing adaptive model selection based on input features, improving handling of missing or low-quality audio and enhancing flexibility over static pipelines. Instead of fixed assumptions, it assesses audio quality in real-time to choose the inference pathway, transforming into an adaptive detection system. This reflects a shift from model-centric to system-centric engineering focused on robustness, adaptability, and deployment.

6.6 Testing

Unit Testing

Unit testing verified individual system components, including frame extraction, audio processing, and model prediction modules, to ensure their correctness and dependability.

Integration Testing

Integration testing was conducted to evaluate the performance of the complete system pipeline:

- Input → preprocessing → model inference → output.

The evaluation focused on data transfer accuracy between pipeline stages, appropriate model selection based on input conditions, and system output consistency and dependability.

Usability Testing

A study with 30 participants assessed the system’s usability, including ease of use, user trust, and output interpretability. Results showed high usability and moderate to high trust. Performance was measured using accuracy, balanced accuracy, ROC-AUC, and MCC. The system was also tested on an external dataset to evaluate generalisation.

6.7 Challenges and Limitations

Multiple challenges arose during development and deployment: class imbalance causing biased outcomes; model instability, especially with Vision Transformer (ViT) architectures; variability in audio quality and accessibility affecting multimodal consistency; and delays in API responses affecting real-time inference. Strategies like multi-dataset training were used to improve class balance and generalisation; refining models for stability; and adjusting thresholds to improve classification accuracy.

6.8 Conclusion

This chapter describes implementing the VerifAI system, transforming design into a functional deepfake detection framework. It combines research with practical development, integrating multiple models, a preprocessing pipeline, and dynamic inference. Testing confirms it meets requirements, with some limitations regarding real-world variability. The next chapter evaluates performance in detail, analysing results, comparing models, and assessing effectiveness.

Chapter 7

Results and Evaluation

This chapter assesses the outcomes of the VerifAI system, presenting experimental results and analysing their significance in relation to the project objectives. It encompasses performance metrics, cross-dataset performance, real-world testing, and user evaluation to provide a comprehensive assessment of effectiveness and usability.

7.1 Introduction

This chapter presents the results of implementing the VerifAI system and evaluates its performance against earlier objectives. It has two main sections: results, which display findings clearly and structurally, and evaluation, which critically assesses these findings. The results focus on key performance metrics, multimodal effectiveness, generalisation, and real-world system behaviour. The evaluation interprets these outcomes, considering strengths, limitations, and overall success within the deepfake detection field.

7.2 Results

7.2.1 Quantitative Model Performance

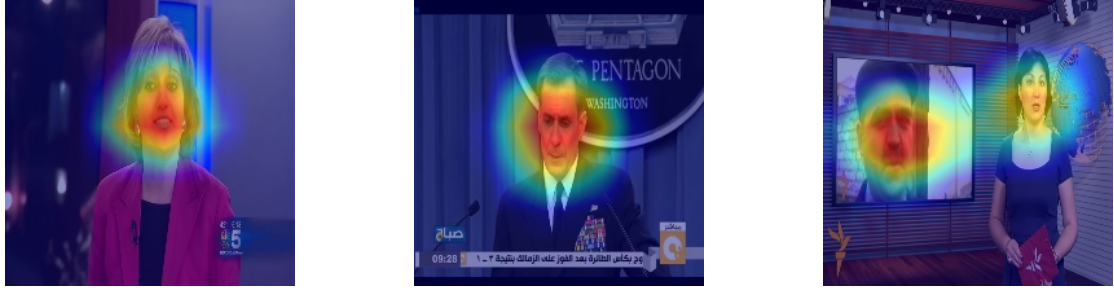
The final VerifAI model was evaluated with various metrics to ensure reliable assessment in balanced and unbalanced scenarios. The system achieved the following performance:

- Accuracy: ≈ 0.99 .
- Balanced Accuracy: ≈ 0.995 .
- ROC-AUC: ≈ 0.996 .
- Average Precision (AP): ≈ 0.999 .
- Matthews Correlation Coefficient (MCC): ≈ 0.84 .

These results demonstrate near-perfect class separation, with high ROC-AUC and balanced accuracy, indicating strong discrimination and consistent performance despite class imbalance.

7.2.2 Multimodal vs Unimodal Performance

The project assessed multimodal versus unimodal detection. Results showed the multimodal model outperformed visual-only and audio-only models and identified cross-modal inconsistencies, like real video with fake audio. Unimodal models struggled with manipulations. Explainability techniques clarified regions influencing predictions, shown in Figure 7.1.



(a) Predicted fake ($p = 66.6\%$) (b) Predicted real ($p = 87.6\%$) (c) Predicted real ($p = 74.6\%$)

Figure 7.1: Example Grad-CAM heatmaps illustrating regions of interest utilised by the model for deepfake detection.

7.2.3 Cross-Dataset Generalisation

The system trained on multiple datasets, including FaceForensics++, Celeb-DF v2, DeepFakeForensics, and FakeAVCeleb, showed consistent cross-dataset performance, reduced overfitting, and increased robustness. Figure 7.2 illustrates false positive rates on real samples across datasets.

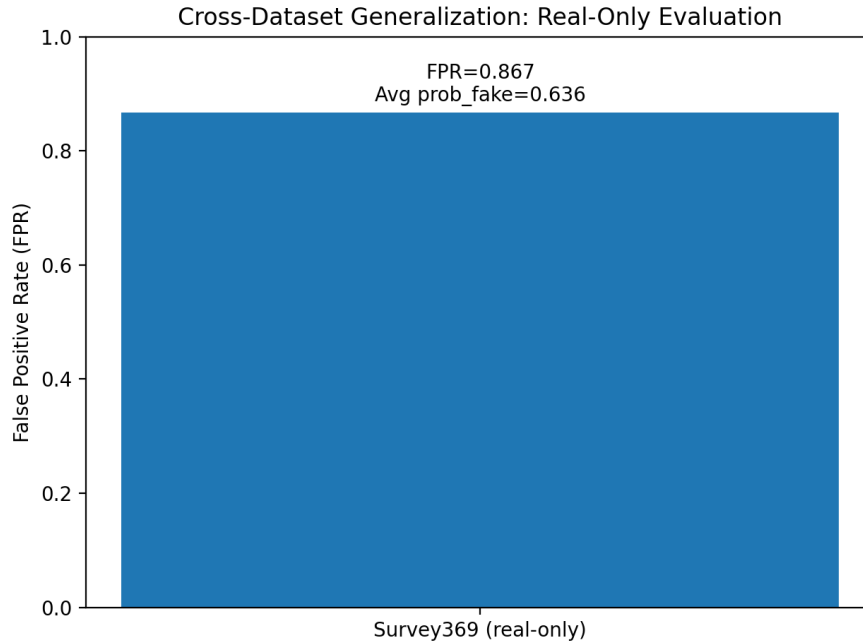


Figure 7.2: Cross-dataset evaluation indicating the false positive rate on real-only samples, emphasising the challenges of generalisation under distribution shift.

7.2.4 Real-World Evaluation

A set of 15 real-world videos, excluded from training and validation, assessed deployment. The model remained robust but less stable than on benchmarks, with variable confidence scores. Lighting, camera quality, and motion-affected detection highlight a distribution shift between curated datasets and real-world inputs.

7.2.5 Inference Stability

Using multi-frame sampling and averaging greatly enhanced prediction stability, reduced noise sensitivity, ensured consistent results, and increased reliability in dynamic video settings.

7.2.6 System Behaviour and Deployment Outputs

The system efficiently generated prediction labels, scores, and inference modes. Dynamic routing worked correctly, using multimodal inference with audio and switching to visual-only fallback when audio was unavailable.

7.2.7 Usability and User Study Results

A user study with 30 participants assessed usability and trust. Participants preferred probability and label outputs over binary results, with confidence scores boosting perceived trustworthiness. A simplified interface improved usability, but some users had trouble interpreting probabilities without help, and explainability features were limited. Figure 7.3 shows response distribution, reflecting overall usability and trust.

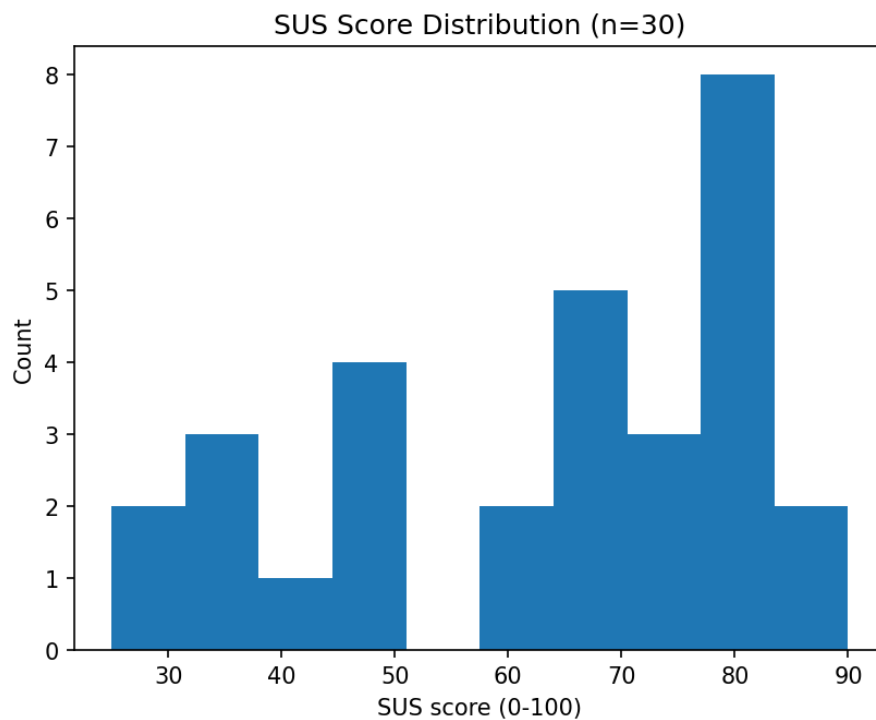


Figure 7.3: Distribution of System Usability Scale (SUS) scores and trust responses from user evaluation.

7.2.8 Results Summary

The results show the multimodal model achieved high accuracy and robustness, detected cross-modal inconsistencies effectively, improved generalisation through multi-dataset training, with more variable real-world performance; the system functioned reliably.

7.3 Evaluation

7.3.1 Evaluation Criteria and Metrics

The system was evaluated using the following metrics:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (7.1)$$

$$BalancedAccuracy = \frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right) \quad (7.2)$$

$$AUC = \int_0^1 TPR(t) d(FPR(t)) \quad (7.3)$$

$$AP = \sum_n (R_n - R_{n-1}) \cdot P_n \quad (7.4)$$

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall} \quad (7.5)$$

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (7.6)$$

Balanced accuracy and MCC were emphasised due to dataset imbalance. Furthermore, the evaluation also considered the capacity for generalisation, multimodal performance, resilience in real-world scenarios, in addition to usability and trustworthiness.

7.3.2 Evaluation Against Objectives

The system successfully met the majority of its objectives:

Objective	Outcome
Multimodal detection	Achieved
Robust pipeline	Achieved
Cross-dataset generalisation	Achieved
Dynamic inference routing	Achieved
Real-world evaluation	Achieved
Deployment as API	Achieved
Explainability	Partially achieved

Table 7.1: Evaluation of system objectives

The only partially achieved objective was explainability, which remained limited compared to the desired level of interpretability.

7.3.3 Strengths of the System

The evaluation highlights several key strengths:

Multimodal Effectiveness The system effectively utilises cross-modal inconsistencies, outperforming unimodal baseline methods.

Dynamic Inference Routing Unlike many existing systems, VerifAI adapts to input conditions, improving robustness.

Generalisation Training on multiple datasets and using subject-level splitting significantly enhances real-world performance.

Stability Multi-frame inference helps diminish prediction variability.

Deployment Readiness The system, based on FastAPI, demonstrates strong practical usability and seamless integration.

7.3.4 Limitations

Compared to existing deepfake detection methods, which typically achieve ROC-AUC scores between 0.90 and 0.98 on benchmark datasets, the VerifAI system shows competitive or sometimes better performance. However, this must be viewed cautiously due to differences in datasets and evaluation protocols. Despite good results, limitations include poor lighting, compression artefacts, and noisy inputs. Multi-dataset training doesn't fully address dataset bias. Limited interpretability lowers non-expert confidence. Multimodal models also need more computational resources than unimodal ones.

7.3.5 Comparison with Existing Work

Feature	Typical Systems	VerifAI
Multimodal	Limited	Yes
Dynamic routing	No	Yes
Real-world testing	Rare	Yes
Explainability	Low	Moderate
Deployment	Limited	Yes

Table 7.2: Comparison with existing deepfake detection systems

This comparison shows VerifAI fills a gap in existing systems by integrating adaptability and multimodal reasoning. Many systems work under static conditions but lack flexibility with varying input modalities. VerifAI's dynamic routing addresses this.

7.3.6 Unexpected Findings

Several unforeseen discoveries emerged: (a) audio-only models proved less reliable than expected, (b) real-world performance was more variable than benchmarks, and (c) clip-level relabelling greatly improved detection accuracy. These highlight the importance of task formulation and evaluation methods.

7.3.7 Methodology Evaluation

The methodology was effective; iterative testing refined the model, subject-level partitioning improved evaluation validity, and practical limitations emerged during real-world testing. Using more real-world data could have strengthened the evaluation, and earlier explainability methods might have helped.

7.4 Conclusion

This chapter critically evaluated the VerifAI system, showing high performance across metrics and addressing key challenges like multimodal integration and generalisation. It confirms the system meets its objectives and improves on traditional deepfake detection but has limitations in real-world robustness and explainability. Although it learns robust features, these depend partly on dataset-specific traits, highlighting the need for domain generalisation. Progress in deepfake detection depends on model complexity and system-level design choices such as multimodal integration, adaptive inference, and realistic evaluation. The next chapter discusses the system’s legal, social, ethical, and professional implications.

Chapter 8

Legal, Social, Ethical and Professional Issues

The development of deepfake detection systems like VerifAI raises legal, social, ethical, and professional issues that must be addressed to ensure the system is effective, responsible, lawful, and aligned with societal expectations. This chapter examines implications related to data protection, user trust, ethics, standards, and potential broader challenges of large-scale deployment.

8.1 Legal and Regulatory Considerations

This project prioritises data protection and privacy, complying with the UK GDPR and Data Protection Act 2018. It involves collecting user-generated video data with informed consent, practising data minimisation, secure storage, and policies for deletion after the project. Participant data was stored securely on University OneDrive, anonymised when possible, and accessed only by the researcher and supervisors, then permanently deleted. Participants were informed of their rights, data use, and research scope. The study avoided sensitive data and limited participation to adults, reducing legal and ethical risks. It used publicly available datasets like FaceForensics++ and FakeAVCeleb under academic licenses, with no redistribution of copyrighted material, ensuring data was for research and education.

8.2 Ethical Considerations

Ethical considerations were central to the design and execution of this project, especially given the sensitive and potentially harmful uses of deepfake technology.

8.2.1 Participant Consent and Privacy

The study followed ethical standards to protect participants and data: participation was voluntary, informed consent was obtained, and no deception or coercion was used. Participants could withdraw anytime without penalty. Data was anonymised and securely stored. No deepfake content was created, reducing misuse and harm.

8.2.2 Risk of Misuse

A key ethical concern is the potential misuse of deepfake detection systems by malicious actors who may use them to enhance deepfake creation, surveillance, or profiling. False results could cause false accusations or harm reputations. To address this, the system offers probability scores instead of definitive labels and is meant to aid decision-making, not replace it.

8.2.3 Bias and Fairness

Dataset bias is an ethical challenge in deepfake detection, as datasets often underrepresent demographics and show a manipulation imbalance. Multi-dataset training helps, but bias remains, raising fairness concerns and inconsistent model performance. Solving this requires larger, diverse datasets and fairness-focused evaluation methods.

8.3 Social Implications

Deepfake technologies have notable societal implications, especially regarding misinformation, trust, and digital media consumption.

8.3.1 Public Awareness and Trust

Survey findings from this project highlight several key societal concerns: 66.94% of participants report awareness of deepfake technologies, 86.45% are concerned about their misuse for misinformation or fraud, and only 22.76% have high confidence in their ability to detect deepfakes. These results reveal a gap between awareness and understanding, emphasising the need for accessible and reliable detection tools.

8.3.2 Trust in AI Detection Systems

The survey reveals limited confidence in AI-based detection tools, with only 32.25% of participants fully trusting them and 44.44% uncertain about their reliability. This underscores a key challenge: technical precision alone isn't enough; building and maintaining user trust is essential. Participants noted that trust is bolstered when systems offer clear, interpretable explanations, confidence scores for context, and transparent, accessible outputs. These findings directly influenced the design of the VerifAI system, especially its focus on explainability and user-friendly features.

8.3.3 Impact on Society

Deepfake tech affects politics, journalism, and personal identity by spreading misinformation, fake news, and enabling fraud and harassment. Survey shows fake news and journalism are top risks (50.41%), highlighting the need for tools like VerifAI for verification, fact-checking, and raising awareness.

8.4 Professional Considerations

From a professional perspective, this project aligns with the ethical and technical principles outlined by recognised bodies such as the British Computer Society (BCS).

8.4.1 Responsibility and Integrity

The development process emphasised responsibility and integrity by transparently reporting results without exaggeration, clearly communicating system limitations, and ethically managing user data. It openly acknowledges uncertainties and boundaries, aligning with best practices in responsible AI development.

8.4.2 Security and Reliability

Various software engineering practices ensured system security and reliability, including input validation to block malicious uploads, secure user data management, and a modular architecture for maintainability and expansion. These measures are vital for safe deployment.

8.4.3 Reproducibility and Research Standards

The project adheres to research standards emphasising reproducibility and transparency. It features structured preprocessing, detailed documentation of training and experiments, and a rigorous evaluation, boosting credibility and aligning with professional standards in software engineering and machine learning.

8.5 Commercial and Deployment Considerations

If VerifAI were to be deployed in a commercial setting, several further technical, legal, and user-focused challenges would need to be addressed.

8.5.1 Scalability and Infrastructure

Commercial deployment requires a scalable infrastructure, including cloud-based architectures to handle extensive video data, optimise latency, and enhance real-time system performance.

8.5.2 Cost Considerations

Deploying multimodal systems involves high costs: computational expenses, storage needs, and ongoing maintenance, updates, and monitoring.

8.5.3 Legal Risks

Commercial deployment raises legal issues: liability from incorrect predictions, misuse risks, and compliance with data laws.

8.5.4 User Trust and Adoption

Survey findings show user trust is a key obstacle; users want transparency and simple explanations. Trust in AI isn't automatic. Successful implementation should focus on explainability, interface usability, and clear communication about limitations and uncertainties.

8.6 Conclusion

This chapter examined the legal, social, ethical, and professional implications of the VerifAI system. While deepfake detection offers societal benefits, it also raises issues like privacy, bias, trust, and misuse. These were addressed through ethical data use, transparent design, and survey-informed user considerations. More research is needed to improve explainability, reduce bias, and promote responsible deployment at scale. The chapter highlights that deepfake detection effectiveness depends on technical performance and adherence to legal, ethical, and societal standards. The next chapter summarises the project, its achievements, and future plans.

Chapter 9

Conclusion

This chapter critically reflects on the VerifAI project, summarising work, evaluating success, and situating it within deepfake detection research and deployment. It highlights key limitations and future directions.

9.1 Summary of the Work

This project tackled detecting deepfake media in real environments amid rapid tech advancements. We developed VerifAI, a multimodal system that combines visual and audio analysis for robust detection. The process included analysing the problem, reviewing literature, defining requirements, designing a modular architecture, implementing an end-to-end pipeline, testing across datasets, and addressing user factors like trust and usability. The key contribution is transitioning from experimental models to a deployable system, bridging research and practical use.

9.2 Evaluation of Outcomes and Objectives

The main goal was to develop a reliable multimodal deepfake detection system for real-world conditions, which was successfully achieved.

9.2.1 Achievement of Objectives

The project achieved its objectives by creating a comprehensive preprocessing pipeline for visual and audio data across datasets, exploring architectures and choosing a deep multimodal fusion model, implementing a dynamic inference routing to handle missing or unreliable audio, adopting a subject-level data split for better evaluation, introducing a clip-level authenticity aligned with real-world detection, and deploying the system as a FastAPI inference service.

9.2.2 Key Technical Contributions

This project introduces several contributions beyond conventional deepfake detection approaches:

- **Dynamic modality-aware inference:** enabling adaptive switching between multimodal and visual-only inference pathways.

- **Deep multimodal fusion:** enhancing representational capacity compared to probability-level fusion methods.
- **Multi-frame inference stabilisation:** reducing prediction variability and improving robustness.
- **Subject-level dataset splitting:** mitigating identity leakage and improving generalisation performance.
- **Clip-level authenticity relabelling:** aligning classification objectives with real-world detection requirements.

These contributions reflect a transition from purely experimental models towards a system designed for practical deployment.

9.2.3 Performance and Effectiveness

The evaluation shows multimodal fusion outperforms unimodal methods and effectively detects cross-modal inconsistencies like audio-only manipulations. Multi-frame aggregation improves prediction stability, and subject-level evaluation offers realistic performance insights. However, challenges remain, such as performance drops with distribution shifts, sensitivity to dataset biases, and variability in real-world scenarios. These findings highlight the need to focus on system robustness, generalisation, and deployment readiness over just model accuracy.

9.3 Broader Significance and Reflection

This project advances the deepfake detection field in two main aspects: bridging the gap between research and deployment, and emphasising the importance of human-centred AI design.

9.3.1 Bridging Research and Deployment

Many approaches perform well in experiments but struggle in real-world scenarios. The VerifAI system demonstrates that robustness depends more on adaptive design than model complexity. Usability improves with transparent outputs and simple deployment, like APIs. Real-world relevance also requires effectively managing incomplete, noisy, or imperfect inputs.

9.3.2 Human-Centred AI Considerations

The project highlights human-centred design in AI, emphasising trust, transparency, and reliability in detection methods. Survey results show users are worried about deepfakes, lack confidence in detecting manipulated media, and need clear explanations and scores. Technical accuracy alone isn't enough; usability and transparency are crucial for real-world trust.

9.3.3 Personal and Technical Reflection

From an engineering perspective, this project revealed that building a deployable system is more complex than training models. Deployment introduces challenges like edge cases, failures, and data variability. Iterative testing is essential to refine models and develop practical solutions. Moving from basic models to a dynamic, multimodal detection framework highlights the importance of system-level thinking in applied machine learning.

9.4 Limitations

Despite the contributions of this project, several limitations should be acknowledged:

- **Dataset limitations:** Existing datasets do not fully capture the variability and complexity of real-world scenarios, limiting generalisation.
- **Limited temporal modelling:** The system relies on frame-based aggregation rather than modelling full temporal sequences, potentially overlooking temporal inconsistencies.
- **Computational constraints:** Resource limitations restricted the exploration and optimisation of more complex architectures.
- **Limited explainability:** While confidence scores are provided, more advanced interpretability methods could enhance transparency and user trust.
- **Scalability considerations:** Although the system is deployable, it has not been optimised for large-scale production environments.

These limitations highlight key areas for future research and system improvement.

9.5 Future Work

Several directions can be explored to further extend and enhance the VerifAI system.

9.5.1 Model and Architecture Improvements

Future research could focus on enhancing the model architecture by incorporating cross-modal attention mechanisms to better link audio and visual features, adopting temporal transformer architectures for more effective sequence modelling, developing improved domain generalisation techniques to increase robustness across datasets, and exploring self-supervised learning methods to lessen reliance on labelled data.

9.5.2 System and Engineering Enhancements

From a systems perspective, several improvements could boost scalability and reliability: deploying within a fully cloud-based infrastructure (such as AWS or containerised environments using Docker), implementing CI/CD pipelines for continuous integration and deployment, optimising system performance for large-scale processing, and enhancing security measures to better withstand adversarial attacks.

9.5.3 Explainability and User Trust

Enhancing explainability is a crucial future goal: integrating Grad-CAM visualisations to identify influential regions, creating intuitive, user-friendly explanation interfaces, and improving the communication of model uncertainty to build trust.

9.5.4 Real-World Deployment

A key next step is transitioning from a research prototype to a fully functional public platform. The system has already been deployed as a production API, with plans for a public release via:

- <https://vrifai.com>

This platform aims to provide real-time deepfake detection, public access for wider use, and integration into actual workflows and applications. Nonetheless, deploying it successfully involves overcoming key challenges: expanding infrastructure to manage large data volumes, ensuring adherence to international legal and regulatory standards, and fostering user trust and adoption.

9.6 Final Remarks

This project shows that effective deepfake detection requires more than just high-performing models. It involves careful system design, thorough evaluation, and consideration of real-world constraints. The VerifAI system advances this by offering a multimodal, adaptive, deployable, and user-aware solution that combines visual and audio data, uses dynamic inference, and has an API setup. As deepfake tech advances, reliable detection becomes even more critical. This work provides a foundation for future research, helping maintain trust in digital media.

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Appendix A

Project Proposal & Plan

Reel or Real? A Context-Aware Deepfake Detection and Verification Framework

1. Introduction & Project Scope

The proliferation of deepfakes – highly realistic synthetic videos generated using advanced AI architectures such as Generative Adversarial Networks (GANs), transformers, and diffusion models – presents a significant threat to digital integrity and public trust. By mimicking the appearance and voices of real individuals, deepfake videos are increasingly used to mislead audiences, manipulate information, and erode institutional credibility.

This project proposes a context-aware deepfake video detection and verification framework. In this context, "context-aware" refers to the system's ability to reason beyond visual signals by integrating multimodal features, metadata analysis, and source verification into the decision process. The system is designed exclusively for detecting AI-generated videos, with future extensibility to image-based deepfakes.

The framework adopts a hybrid architecture of convolutional neural networks (CNNs) and transformers to model both spatial and temporal patterns in deepfake videos. It incorporates explainable AI techniques such as heatmaps and attention maps, a user-friendly interface to support transparency, and cryptographic watermarking for provenance verification. The project aims to deliver a detection tool that is robust, transparent, and ethically grounded, and suitable for deployment in real-world environments.

2. Rationale & Background Research

While deepfake detection has received growing academic attention, most systems still suffer from poor generalisation, limited multimodal integration, and low interpretability. CNN-based detectors like Xception and ResNet (Pan et al., 2020; Ahmed et al., 2022) perform well on

curated datasets but often fail on unseen or real-world content. Transformer-based models (Gong & Li, 2024; Zhang et al., 2025) offer improved temporal modelling but remain black-box in nature, limiting their transparency and trustworthiness.

Dataset quality also remains a major factor. Early benchmarks such as FaceForensics++ and the Deepfake Detection Challenge lack diversity and real-world complexity (Lyu, 2020). In contrast, more recent datasets like KoDF, HAV-DF, and WildDeepfake support better generalisation across demographics, while FakeAVCeleb introduces complex audio-visual scenarios for multimodal learning (Khalid et al., 2021).

Research has also begun to explore provenance-based validation. Standards like C2PA (Laskar, 2025) embed verifiable metadata, while cryptographic watermarking (Walczyna et al., 2025; Wang et al., 2025) offers tamper-evident content tracking. However, such methods are rarely integrated into deep learning pipelines.

Key gaps persist as models often generalise poorly across datasets; audio-visual fusion remains underdeveloped; explainability features are lacking; provenance tools are seldom embedded; and user-centric design is rarely prioritised. This project addresses these challenges by developing a unified, explainable, and ethically responsible deepfake video detection framework, grounded in multimodal learning and real-world usability.

3. Project Development: Approach & Evaluation

The proposed system will follow a modular hybrid architecture that combines CNNs and transformers. The CNN backbone will extract spatial features from individual video frames, while the transformer component will model temporal dependencies across sequences. To incorporate audio information, the system will use Mel-frequency cepstral coefficients (MFCCs), allowing for the alignment of audio

patterns with their corresponding visual frames. This multimodal approach is intended to detect inconsistencies in both synchronisation and content, which are often indicative of deepfake manipulations.

The framework will be trained and evaluated using a focused selection of publicly available deepfake video datasets. For core implementation, three datasets will be used: the Deepfake Detection Challenge (DFDC) for large-scale and diverse video content, FakeAVCeleb for challenging audio-visual fusion scenarios, and KoDF (Korean Deepfake Dataset) to ensure linguistic and demographic diversity. Together, these datasets provide a robust foundation for training and evaluating multimodal models. Additionally, a user-contributed dataset, containing videos, audio, and images from over 350 participants, may be incorporated pending university ethical approval, offering real-world variability and authenticity.

While datasets such as FaceForensics++, HAV-DF, and WildDeepfake have also been reviewed during the literature phase, they are not part of the current implementation plan. However, they remain viable for future evaluation and extension, particularly in exploring cross-domain generalisation and static image detection. The framework is designed to accommodate such extensions as part of ongoing research beyond the initial project scope.

To enhance transparency and user trust, the system will embed explainable artificial intelligence mechanisms directly into its inference pipeline. Grad-CAM heatmaps will be used to highlight areas of visual interest, while transformer-based attention maps will offer insight into how sequential information contributes to classification decisions. Confidence scoring will be displayed alongside each output, quantifying the model's certainty. Furthermore, the framework will integrate metadata verification aligned with the Coalition for Content Provenance and Authenticity (C2PA) standard, allowing users to assess the source

integrity of submitted content. These explainability and provenance tools will be presented through a user-facing interface designed for accessibility and interpretability.

Model evaluation will involve both quantitative and qualitative methods. Quantitative assessment will include standard metrics such as accuracy, precision, recall, F1-score, and robustness against compression or distortion. These will be benchmarked across DFDC, FakeAVCeleb, and KoDF to assess generalisability. Qualitative evaluation will be conducted through a small user study, focusing on how clearly the system communicates its decisions and how much trust users place in its outputs.

4. Feasibility, Planning & Gantt Outline

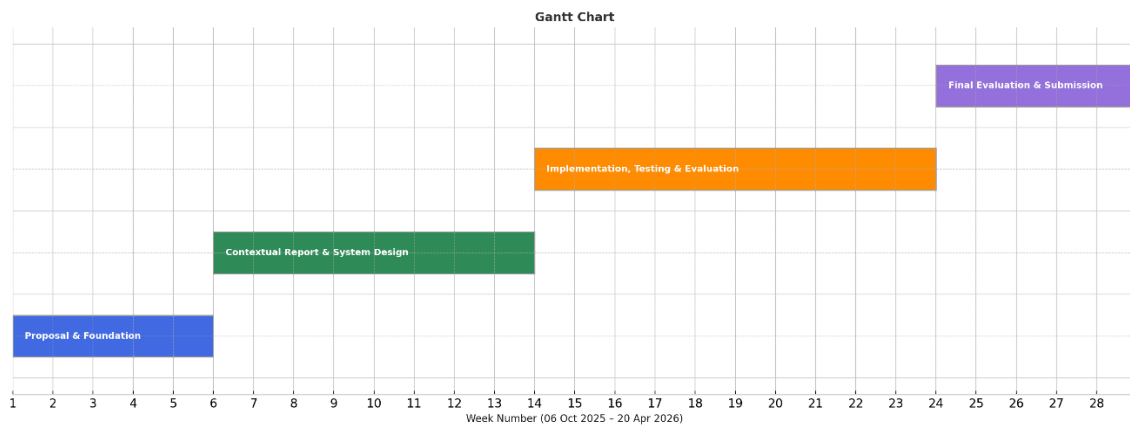
The system will be developed using Python, leveraging PyTorch, TensorFlow, and OpenCV as the primary libraries for model implementation and data handling. A GPU-supported laptop is available to support local training, while scalable cloud computing platforms such as Google Colab Pro may be used to accommodate more computationally intensive tasks. The architecture will be designed exclusively for detection and verification purposes; no component of the system will be capable of generating synthetic content. This design choice ensures alignment with responsible AI practices and reduces ethical and legal risks associated with content creation.

To ensure ethical compliance, all datasets employed in the project will either be openly licensed or approved by a university ethics committee. The user-contributed dataset, sourced from over 350 participants, may be used only upon formal university ethical approval. Participants will provide informed consent, and all submissions will be anonymised and stored securely. The system will operate in accordance with the General Data Protection Regulation (GDPR) and institutional ethical guidelines,

including routine audits for fairness and demographic bias throughout the development lifecycle.

<i>Task</i>	<i>Duration</i>	<i>Key Activities</i>	<i>Exp. Outputs</i>
Proposal & Foundation	Weeks 1-5 (Oct-Nov)	Define project scope, conduct in-depth literature review, and analyse available datasets (e.g., FaceForensics++, HAV-DF, WildDeepfake) for background context. Finalise core datasets for implementation (DFDC, FakeAVCeleb, KoDF) and prepare project proposal.	Project scope statement, annotated bibliography, proposal submission.
Contextual Report & System Design	Weeks 6-13 (Nov-Jan)	Submit ethics documentation for survey dataset. Conduct detailed system design using UML and architectural diagrams. Implement and evaluate a baseline CNN model; compile contextual report.	Ethics submission, architecture diagrams, contextual report.
Implementation, Testing & Evaluation	Weeks 14-23 (Jan-Mar)	Build full data-processing pipeline; train and evaluate CNN, transformer, and multimodal fusion models. Conduct explainability integration and user studies to assess trust and usability.	Trained models, evaluation results, explainability visuals, user survey analysis.
Final Evaluation & Submission	Weeks 24-28 (Mar-Apr)	Consolidate quantitative and qualitative data. Write and proofread final report, prepare presentation slides, final demo, and submit all deliverables.	Final report, presentation, code submission, and project demonstration.

Potential risks include dataset imbalance, limited computational resources, and overfitting. Mitigation strategies include data augmentation, cloud-based training, and early stopping. The project's structured timeline and modular design ensure successful completion within the academic year.



5. References

<i>Paper</i>	<i>Objective</i>	<i>Algorithm</i>	<i>Dataset</i>	<i>Methodology</i>	<i>Evaluation & Findings</i>
Ahmed et al. (2022)	Survey CNN-based detection methods	CNNs (VGG, ResNet)	FaceForensics++, DFDC	Comparative literature review	Accuracy, precision - CNNs perform well but generalise poorly
Gong & Li (2024)	Review detection algorithms and datasets	CNNs, Transformers	KoDF, DFDC, WildDeepfake	Broad SoTA review	Mixed metrics - Transformers generalise better but are less interpretable
Heidari et al. (2023)	Systematic review of DL-based detection	CNNs, GANs, RNNs	DFDC, FaceForensics++	Categorisation and taxonomy	Varied metrics - Hybrid models outperform single CNNs
Lyu (2020)	Discuss detection challenges and next steps	Overview	N/A	Conceptual/position paper	No metrics - Highlights trust, robustness, and generalisation issues
Khalid et al. (2021)	Release FakeAVCeleb multimodal dataset	CNN + audio alignment	FakeAVCeleb	Dataset creation + baseline evaluation	AUC - Multimodal fusion improves accuracy

Kaur et al. (2024)	Develop Hindi A/V deepfake dataset	Audio-video fusion	HAV-DF	Synthetic data generation	AUC, F1 - Hindi deepfakes pose distinct challenges
Kwon et al. (2021)	Create Korean deepfake dataset	Deepfake gen. + CNN	KoDF	Dataset curation + benchmarking	Accuracy - Supports model generalisation across ethnic groups
Zi et al. (2020)	Release real-world video dataset	Benchmark eval	WildDeepfake	Web scraping + annotation	Accuracy - High variability, useful for generalisation
Pan et al. (2020)	Test CNN for video detection	Xception	FaceForensics++	Model training + testing	Accuracy - High on benchmark data but limited generalisation
Laskar (2025)	Explore ethical provenance tracking	Metadata (C2PA)	N/A	Technical + ethical review	N/A - Provenance enhances trust in source media
Walczyna et al. (2025)	Undo deepfake swaps via watermarking	RE-Mark	Synthetic face-swaps	Watermark embed + recovery	Identity recovery rate - Effectively restores original identity
Wang et al. (2025)	Initiative-taking tamper detection	FractalForensics	Synthetic videos	Watermark injection + detection	Precision, recall - Accurately localises manipulations
Zhang et al. (2025)	Build robust multimodal detection model	MFCLIP (Transformer + CLIP)	Diffusion Face Dataset	Contrastive learning + fusion	F1-score - Excellent on unseen forgery types

WEEKLY PROJECT PLAN (06 Oct 2025 – 20 Apr 2026)

PHASE 1 — Proposal & Foundation (6 Oct – 10 Nov 2025)

Goal: Establish theoretical foundation, validate research gap, and complete proposal submission.

Week	Dates	Focus & Tasks	Expected Outputs
W1	6–12 Oct	Define final project scope and aims (framework, not standalone app). Conduct a deep literature overview using Heidari et al. (2023) and Gong & Li (2024) to map detection methods (CNN, Transformer, multimodal, biological). Identify 3–4 subproblems: dataset bias, multimodal fusion, explainability, real-time deployment.	Project scope statement, subproblem table, annotated bibliography (10+ key papers).
W2	13–19 Oct	Detailed review of datasets (FaceForensics++, DFDC, KoDF, WildDeepfake, FakeAVCeleb). Compare dataset properties, diversity, realism (Kwon et al. 2021; Zi et al. 2020; Khalid et al. 2021). Begin initial draft of Problem Domain section.	Dataset comparison matrix, problem domain draft.
W3	20–26 Oct	Research baseline models from Pan et al. (2020) and Ahmed et al. (2022). Sketch high-level architecture: CNN + transformer hybrid. Draft Methodology and Evaluation Plan sections.	Initial proposal draft, architecture diagram.
W4	27 Oct – 2 Nov	Finalize Introduction, Problem Domain, Methodology, and Evaluation. Proofread and format proposal.	Proposal v2 for supervisor feedback.
W5	3–9 Nov	Incorporate supervisor comments. Final checks: citations (Harvard), references, ethics summary.	Final Proposal

PHASE 2 — Contextual Report & System Design (10 Nov 2025 – 12 Jan 2026)

Goal: Deepen research, design the system architecture, complete contextual report.

Week	Dates	Focus & Tasks	Expected Outputs
W6	10–16 Nov	Prepare full ethics pack (PIS, consent forms, questionnaire). Submit to Faculty Ethics Committee.	Ethics submission.
W7	17–23 Nov	Literature deep-dive: dataset realism & cross-dataset generalization issues (Kwon et al., Zi	Contextual Report draft –

		et al.). Start contextual report (literature review section).	Section 1 (Intro & Lit Review).
W8	24–30 Nov	Requirements engineering: functional, non-functional, ethical requirements. Use UML (use cases, class diagrams).	Requirements specification document.
W9	1–7 Dec	Design detailed system architecture: modular backend, detection module, frontend UI. Define APIs and data flow diagrams. Reference Lyu (2020) for explainability design.	Design diagrams and architecture specification.
W10	8–14 Dec	Implement small baseline CNN model (VGG-Face or MobileNet) on FaceForensics++ subset (Pan et al. 2020). Document experiment results.	Baseline model test report.
W11	15–21 Dec	Draft Methodology and Implementation Plan sections of contextual report. Incorporate journal-based justification (Ahmed et al. 2022; Heidari et al. 2023).	Contextual Report v1.
W12	22–28 Dec	Light week (holiday buffer). Format contextual report and prepare for submission.	Polished draft.
W13	29 Dec – 12 Jan	Final proofread and supervisor review. Integrate all diagrams, tables, references.	Submit Contextual Report.

PHASE 3 — Implementation, Testing & Evaluation (13 Jan – 22 Mar 2026)

Goal: Build, test, and refine the detection framework and evaluation pipeline.

Week	Dates	Focus & Tasks	Expected Outputs
W14	13–19 Jan	Build full data-preprocessing pipeline: frame extraction, audio segmentation, MFCC feature generation. Dataset integration (KoDF + FakeAVCeleb).	Preprocessing scripts, verified data samples.
W15	20–26 Jan	Train and evaluate visual CNN (Xception / MobileNet). Measure dataset-specific performance (FaceForensics++).	Visual model results (Accuracy, F1, AUC).
W16	27 Jan – 2 Feb	Implement audio branch using MFCC features and ResNet model (Khalid et al. 2021). Evaluate audio-only performance.	Audio model results, confusion matrix.
W17	3–9 Feb	Implement fusion (multi-modal) detector combining audio and video features. Begin API integration.	Multimodal model v1, feature fusion logs.

W18	10–16 Feb	Conduct internal testing, optimize parameters. Begin pilot usability study with 5 users. Collect qualitative feedback.	Pilot usability notes, model tuning.
W19	17–23 Feb	Apply explainability (Grad-CAM) for heatmaps to visualize fake regions (Lyu 2020). Evaluate cross-dataset generalization (KoDF → WildDeepfake).	Explainability visuals, cross-test accuracy chart.
W20	24 Feb – 2 Mar	Conduct main user study (~30 participants). Collect SUS data and user trust metrics (Sippy et al. 2024).	Raw user data, anonymized survey responses.
W21	3–9 Mar	Analyse survey results: mean SUS, qualitative comments. Statistical analysis (t-tests, mean comparisons).	Evaluation tables and graphs.
W22	10–16 Mar	Final optimization and code refactoring. Document APIs, backend integration, and system testing.	Finalized framework and documentation.
W23	17–23 Mar	Supervisor demo; fix minor bugs and performance issues.	Stable build v2.0.

PHASE 4 — Evaluation, Final Report & Submission (24 Mar – 20 Apr 2026)

Goal: Consolidate results, finalize report, prepare for submission and presentation.

Week	Dates	Focus & Tasks	Expected Outputs
W24	24–30 Mar	Compile evaluation data (accuracy, F1, SUS, latency). Generate visual results: ROC curves, heatmaps, confusion matrices.	Evaluation visuals and tables.
W25	31 Mar – 6 Apr	Write Results and Evaluation sections of final report. Integrate findings with literature discussion (link to Gong & Li 2024; Heidari 2023).	Report draft v1.
W26	7–13 Apr	Write Discussion, Conclusion, Future Work. Proofread for clarity and structure.	Full report v2.
W27	14–17 Apr	Prepare final presentation slides and demo. Final supervisor review and revisions.	Final report + presentation ready.
W28	18–19 Apr	Submit final report, source code, and demo video. Backup all materials.	Submission completed.
W29	20 Apr	Final deadline – Project submission day	Project officially submitted.

Appendix B

Technical Design

KoDF Usage Note (Week 19)

Background

KoDF (Korean Deepfake Detection Dataset) is a large-scale deepfake dataset introduced by Kwon et al. (ICCV 2021), containing over 60,000 real videos and 175,000 synthesized videos generated using multiple deepfake methods.

Intended Role in This Project

KoDF is referenced as a **source training dataset** to motivate cross-dataset generalization experiments. Models used in this project were originally trained using KoDF-style data distributions, following established deepfake detection literature.

Practical Constraint

The full KoDF dataset exceeds **500 GB uncompressed**. Due to hardware and storage limitations on a personal laptop, the complete video corpus was not re-downloaded or re-extracted during Week 19.

Experimental Decision

Instead of re-ingesting KoDF locally, **cross-dataset generalization** was evaluated by testing trained models on Celeb-DF-v2, a standard unseen-domain benchmark.

Reference

Kwon, P. et al., **KoDF: A Large-Scale Korean DeepFake Detection Dataset**, ICCV 2021.

Appendix C

Experimental Work & Model Development

Baseline CNN Experiment Report

FaceForensics++ (C23) - MobileNetV2

Objective

The objective of this experiment is to establish a lightweight baseline deepfake detection model against which more advanced multimodal and temporal approaches can later be compared. A frame-based Convolutional Neural Network (CNN) is used to assess whether visual information alone is sufficient for reliable deepfake detection on the FaceForensics++ (C23) dataset.

Dataset and Preprocessing

- Dataset: FaceForensics++ (C23 compression)
- Classes:
 - Real (original videos)
 - Fake (DeepFakes, Face2Face, FaceSwap, FaceShifter, NeuralTextures)

The dataset was split at the video level to avoid data leakage:

Split	Videos
Train	4,200
Validation	900
Test	900

Only videos with successfully extracted frames were included in training and evaluation.

To reduce storage and computational cost while maintaining representativeness:

- 5 frames per video were extracted
- Frames were uniformly sampled across the video duration
- Total frames used:
 - Train: 8,500
 - Validation: 3,555
 - Test: 3,560 (approx.)

This approach aligns with common baseline strategies reported in deepfake detection literature.

Model Architecture

- Backbone: MobileNetV2 (pretrained on ImageNet)

- Modification: Final classifier replaced with a single-output linear layer for binary classification
- Output: One logit per frame (fake vs real)
- Loss Function: Binary Cross-Entropy with Logits (BCEWithLogitsLoss)
- Optimizer: AdamW
- Learning Rate: 1×10^{-4}
- Epochs: 5
- Hardware: CPU

This architecture was intentionally kept simple to serve as a baseline rather than a state-of-the-art detector.

Evaluation Methodology

Since the model operates at the frame level, video-level predictions were obtained by:

$$\hat{y}_{video} = \frac{1}{N} \sum_{i=1}^N \sigma(z_i)$$

Where:

- z_i is the logit output for frame i
- $\sigma(\cdot)$ is the sigmoid function
- $N=5$ frames per video

A video was classified as fake if:

$$\hat{y}_{video} \geq 0.5$$

Results

Test Set Performance (Video-Level)

- Number of test videos: 712
- Accuracy: 73.7%
- ROC-AUC: 0.53

Confusion Matrix

$$\begin{bmatrix} 18 & 146 \\ 41 & 507 \end{bmatrix}$$

Ground Truth	Predicted Real	Predicted Fake
Real	18	146
Fake	41	507

Classification Report

Class	Precision	Recall	F1-Score	Support
Real (0)	0.305	0.110	0.161	164
Fake (1)	0.776	0.925	0.844	548
Accuracy			0.737	712

Analysis and Discussion

The baseline CNN demonstrates strong performance in detecting manipulated videos, achieving a fake-class recall of 92.5%, indicating that most deepfake videos are correctly identified.

However, performance on real videos is notably weaker, with a recall of only 10.9%, revealing a strong bias toward the fake class. This behaviour is consistent with prior research on FaceForensics++, where class imbalance and visually subtle compression artefacts often cause models to over-predict manipulation.

The relatively low ROC-AUC score (≈ 0.53) suggests limited ranking capability across confidence thresholds. This is expected given:

- the absence of temporal modelling
- the use of only five frames per video
- lack of facial alignment or landmark tracking
- and the exclusion of audio and metadata cues

Despite these limitations, the achieved accuracy of 73.7% confirms that even a simple frame-based CNN can capture meaningful visual artefacts associated with deepfake generation.

Conclusion

This experiment establishes a valid and reproducible baseline for deepfake detection using a lightweight CNN. The results highlight both the strengths and limitations of purely visual, frame-based approaches and motivate the need for:

- temporal modelling
- multimodal fusion (audio + video)
- and provenance-aware analysis

Skeleton Project Report

1. Reading this week – Journal Log Interview

Momin et al. (2025) present a comprehensive survey of explainable deepfake detection systems across visual, audio, and textual domains. The study highlights that although current detectors achieve high accuracy, most remain opaque and difficult to trust. The authors categorise explainability techniques such as Grad-CAM and attention visualisation and emphasise the necessity of embedding transparency directly into detection pipelines rather than treating it as a post-hoc feature.

Haq et al. (2023) introduce a neuro-symbolic multimodal approach to deepfake detection that leverages emotional inconsistencies between facial expressions and vocal tone. By combining deep learning with symbolic reasoning, the framework detects mismatched affective cues that frequently occur in synthetic media. The work demonstrates that emotionally grounded cross-modal reasoning can significantly improve robustness and interpretability.

Qureshi et al. (2024) provide a large-scale survey of multimodal media forensics, reviewing detection techniques for images, video, and audio on social platforms. The study identifies critical challenges, including cross-dataset generalisation, dataset bias, and the lack of unified multimodal benchmarks. The authors conclude that future detectors must operate across modalities to remain effective against evolving deepfake generation techniques.

Wang et al. (2025) propose ERF-BA-TFD+, a state-of-the-art audio-visual deepfake detector combining enhanced convolutional receptive fields with transformer-based temporal modelling. The method achieves top performance on recent audio-visual benchmarks, demonstrating the effectiveness of joint temporal and cross-modal feature modelling.

Finally, Sippy et al. (2024) examine public trust and perception of deepfakes through a large UK-based survey. The findings show high public concern and low confidence in identifying synthetic media, reinforcing the importance of detection systems that prioritise clarity, transparency, and user trust alongside technical accuracy.

2. Skeleton Project Report Structure

2.1 Abstract

The project investigates the development of a context-aware deepfake video detection and verification framework designed to detect AI-generated videos with high robustness and transparency. The proposed system integrates visual, audio, and provenance-based signals using a hybrid CNN-Transformer architecture. Explainable AI mechanisms and metadata verification are incorporated to improve interpretability and user trust. Evaluation is conducted using standard performance metrics and qualitative user feedback.

2.2 Introduction

The rapid advancement of generative AI has enabled the creation of highly realistic deepfake videos, posing severe threats to information integrity, public trust, and digital security. Existing detection methods primarily rely on visual artefacts and often fail when applied to unseen or real-world data.

This project addresses these limitations by proposing a context-aware detection framework capable of reasoning beyond visual features. By incorporating multimodal learning, explainability, and provenance verification, the framework aims to improve robustness, transparency, and real-world applicability.

2.3 Literature Review

Existing deepfake detectors are dominated by CNN-based approaches that perform well on benchmark datasets but generalise poorly. Transformer-based models improve temporal reasoning but introduce interpretability challenges. Recent research demonstrates the effectiveness of multimodal detection by aligning audio and visual signals, yet full integration remains limited.

Explainability remains underdeveloped, with most systems functioning as black boxes. In parallel, provenance-based solutions such as metadata authentication and watermarking show promise but are rarely combined with AI-driven detection.

The literature identifies five persistent gaps: poor generalisation, weak audio-visual fusion, limited explainability, lack of provenance integration, and minimal

user-centric evaluation. The proposed framework is designed to address these gaps in a unified manner.

2.4 Methodology

The framework adopts a hybrid CNN-Transformer architecture. A convolutional backbone extracts spatial features from video frames, while a transformer module models temporal dependencies across frame sequences. Audio features are extracted using MFCCs and aligned temporally with visual data.

Multimodal fusion combines audio and visual representations to detect synchronisation inconsistencies. Explainable AI techniques, including Grad-CAM heatmaps and attention visualisations, are embedded within the inference process. Metadata and provenance checks aligned with C2PA standards complement the detection pipeline.

2.5 System Design

The system is designed as a modular architecture comprising preprocessing, visual analysis, audio analysis, fusion, explainability, and verification modules. Clear separation of components ensures scalability, maintainability, and extensibility.

Data flows from video ingestion through preprocessing and model inference to output generation, which includes prediction scores, visual explanations, and provenance indicators. UML diagrams and architectural schematics are used to formalise the design.

2.6 Implementation Plan

The system is implemented using Python, leveraging PyTorch, OpenCV, and audio processing libraries. Core datasets include DFDC, FakeAVCeleb, and KoDF. Development follows a staged approach: baseline model implementation, multimodal integration, explainability embedding, and system consolidation.

GPU-supported local training is supplemented with cloud resources where required. Model validation occurs iteratively throughout development to minimise overfitting and ensure stability.

2.7 Evaluation Strategy

Quantitative evaluation employs accuracy, precision, recall, F1-score, and AUC across multiple datasets. Robustness testing includes compression and cross-dataset validation.

Qualitative evaluation is conducted through a user study assessing system clarity, trust, and usability using structured questionnaires. The combined evaluation approach ensures both technical effectiveness and user-centred reliability.

2.8 Ethical Considerations

All datasets are either officially licensed or ethically approved. Any user-contributed data is collected with informed consent and anonymised in accordance with GDPR guidelines.

The framework is limited to detection and verification, with no capability for content generation. Bias assessment and transparency are prioritised to minimise ethical and societal risks associated with automated deepfake judgments.

2.9 Future Work and Limitations

Current limitations include dataset dependency and computational constraints. Future work may extend the framework to image-based deepfakes, real-time detection, adversarial robustness, and expanded provenance analysis.

Further improvements may include enhanced user interfaces, larger-scale user studies, and integration with content moderation pipelines.

Appendix D

User Study & Ethics

User Study Report - Usability & Trust Evaluation

The objective of this user study was to evaluate the system usability of the VerifAI deepfake detection interface using the System Usability Scale (SUS), as well as to assess user trust and confidence in AI-generated deepfake predictions, informed by trust-related metrics described in Sippy et al. (2024). This study was conducted as part of the human-centred evaluation phase of the VerifAI framework, complementing prior technical and cross-dataset performance analyses.

The study involved a total of 30 participants who interacted with the VerifAI system over two consecutive days, with 15 participants completing the study on 31 January 2026 and the remaining 15 on 1 February 2026. Participants were grouped into two age ranges, 18-30 and 31-60, and all sessions were conducted using my personal laptop to ensure consistency in interaction conditions. Ethical considerations were addressed through the provision of a participant information sheet, informed consent form, task instructions, and a post-study debrief. All collected data was anonymised, and no personally identifiable information was stored.

Participants were asked to complete a predefined interaction task involving the use of the VerifAI system to inspect deepfake detection outputs and model confidence scores. During the task, interaction metrics such as task completion time, task success, and error counts were recorded. Following task completion, participants completed a post-task questionnaire consisting of the ten SUS items and additional questions related to trust in the system, perceived reliability, and prior awareness of deepfake technology.

The collected responses were processed using an automated analysis script that computed individual SUS scores on a 0-100 scale, along with an aggregated trust index derived from user-reported confidence measures. The resulting SUS score distribution indicated a wide but realistic range of usability perceptions, with most participants scoring the system within the “good usability” range (70-80). Lower scores were associated with longer task completion times and higher error counts, while higher scores corresponded to faster task completion and fewer interaction issues. This variation reflects expected differences in user familiarity and technical confidence.

Trust-related results showed moderate to high confidence in the system’s predictions for the majority of participants, although trust levels varied independently from usability scores in some cases. This finding aligns with existing research suggesting that system usability and trust, while related, represent distinct dimensions of user experience. Overall, participants demonstrated an ability to engage with the system effectively, and the results suggest that VerifAI provides a usable and reasonably trustworthy interface for deepfake detection tasks.

In conclusion, this user study provides evidence that the VerifAI framework achieves satisfactory usability and user trust in a realistic interaction setting. These findings support the feasibility of the system for practical use and highlight the importance of combining technical performance evaluation with human-centred assessment. The results obtained in this study form a valuable foundation for future, larger-scale user evaluations and further refinement of the system's interface and explainability features.

Appendix E

Technical Documentation & System Details

How to Use Open the live demo: <https://verifai-backend-z6tz.onrender.com>

1. Click **Select File**
2. Choose a short video file:
 - .mp4
 - .mov
 - .avi

The video might include audio, lack audio, or feature fake video, fake audio, or both.

3. Click **Detect**
4. Wait a few seconds while the system processes the upload
5. View the result

Output Format Example response:

```
{
  "label": "fake",
  "prob_fake": "93.41",
  "mode": "hybrid_av"
}
```

Output Explanation

label

- "real" → predicted authentic
- "fake" → predicted manipulated

prob_fake Probability (in %) that the video is fake.
Example: 93.41 means **93.41% confidence** the video is fake.

mode Indicates which detection pipeline was used:

- **hybrid_av** → Multimodal fusion model (video + audio)
- **visual_only_fallback** → Vision-only fallback model

The system automatically selects the appropriate model based on audio presence.

System Architecture

Inference Pipeline

Extract five representative video frames, extract the audio track if available, compute MFCC audio features, and determine whether the audio contains a usable signal. Route the data accordingly: if both video and audio are present, use the Multimodal Fusion Model; if only video is available, fallback to the Vision Transformer (ViT).

Multimodal Fusion Model

Video Branch MobileNetV2 backbone (pretrained on ImageNet), frame-level embedding, with temporal averaging across frames.

Audio Branch ResNet18 backbone adapted for 1-channel MFCC input.

Fusion Concatenating video and audio embeddings, followed by classification using an MLP, with BCEWithLogitsLoss and Sigmoid activation to produce probability scores.

Visual Fallback Model

Vision Transformer (ViT), multi-frame scoring, averaging logits across frames, and stable probability estimation.

Technology Stack

Backend

- FastAPI
- Uvicorn

Deep Learning

- PyTorch
- Torchvision
- timm (Vision Transformer)

Audio Processing

- Librosa
- MFCC feature extraction

Video Processing

- OpenCV
- MoviePy

Deployment

- Render cloud hosting
- Production-ready FastAPI service

Project Structure

```
VerifAI/  
    backend/  
    ml/  
    pipelines/  
    data/  
    experiments/  
    scripts/  
    docs/  
    venv/
```